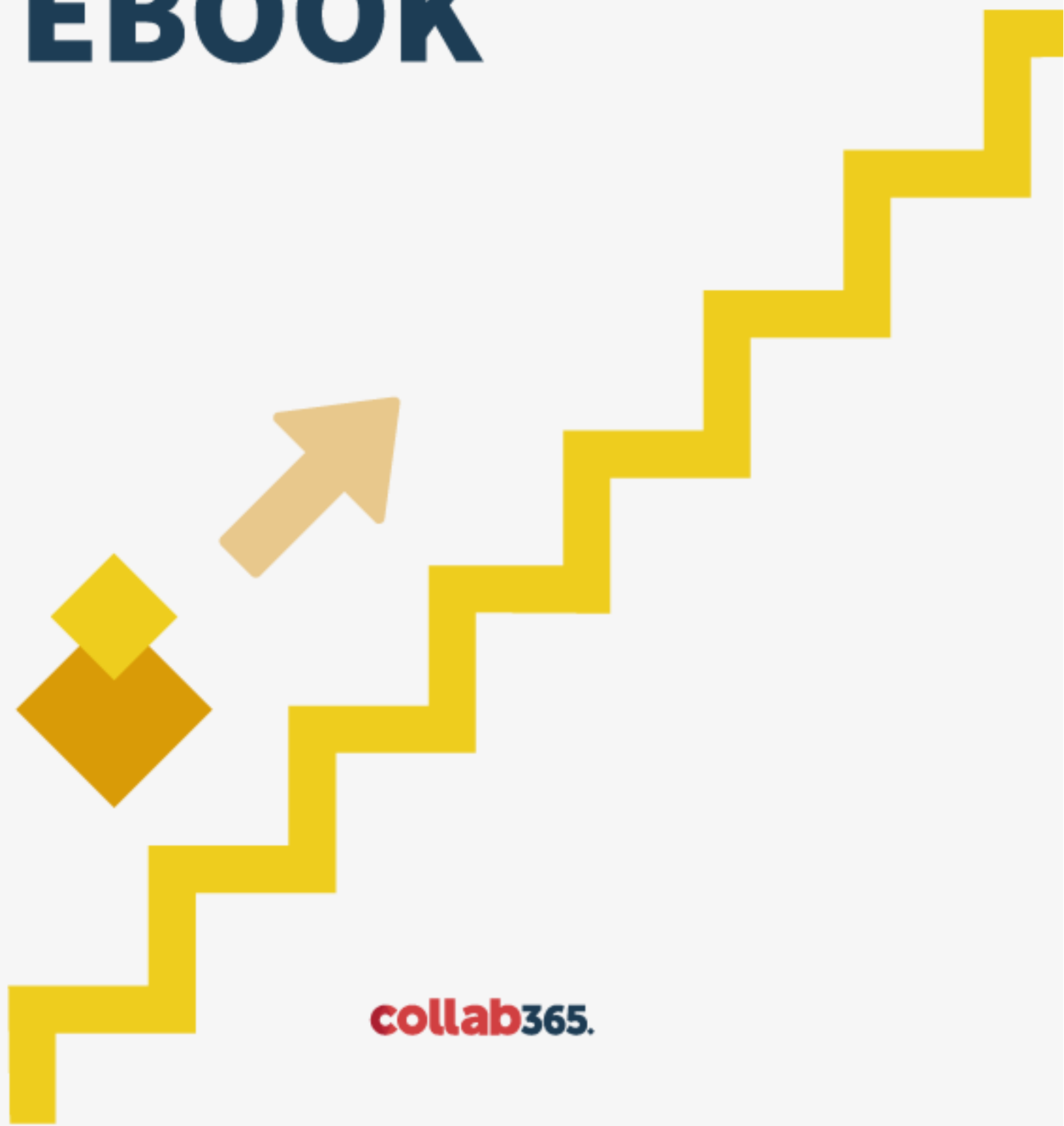


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BEGINNER EBOOK



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Beginners Guide to Power BI

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Introduction

We're living in the era of where data is everywhere. Most of our movements online are noted, measured or categorised. We offer up data about ourselves in return for discounts and access to services.

Those same services and organisations collect data about their own activities and operations. They track efficiency and measure results at every level of business.

It's not just large corporations. Small companies, individuals and groups collect and use this data to reveal insights and help drive experiments, adaptations and behaviours.

There is every reason to care about data as a part of our day to day lives and even more to explore how we can use it to benefit us.

We store data in many forms. Sometimes it's just notes. In business, it's often Excel files, or we might save it to SharePoint lists; it could be in a SQL or an Access database. Wherever it is, typically, it shares one thing in common: it's unconnected and siloed.

This is fine if we don't have a need to analyse or attempt to gain insights from this data. If it's just casual collection (say a list of items for an upcoming trip or presentation) the use tends to be personal.

However, if we collect data because we want to use it to help reveal insights into behaviour or activities taking place – across several different points where we capture the information, having a joined-up view becomes more important.

This is where Power BI (business intelligence) comes in. A Microsoft product which integrates into the Microsoft 365 range of apps, it brings together all your data into handy dashboards, data models and reports.

Ultimately, it's main purpose is to allow you to analyse your data (from one or many sources) while presenting it in a visually meaningful way. And by connecting and visualising your data, individuals, teams and organisations can gain great insight into patterns and trends. Crucially, you don't have to be a data scientist or a programmer to use Power BI.

While you don't have to be a data scientist or a programmer to follow this eBook, it would help if you had some Microsoft 365 experience and/or with the following:

- Familiar with the way data files can be stored and shared - such as in OneDrive / SharePoint or in Teams.
- As we will be exporting data to SharePoint online, knowledge of how to edit a SharePoint page would help (but isn't essential)
- A basic understanding column in Excel along with cells and types of data - such as number, currency etc. but don't worry – you won't see any code or come across any tricky Excel formulae!
- A broad familiarity with what Microsoft 365 offers in general as the Power BI service sits neatly within this ecosystem

To get the most out of this eBook be sure to sign up to a trial version of the Power BI service and to download the Power BI Desktop app. (more on how to do this later on)

Before you begin, we also want to mention that Power BI as a topic and a skill set is **large**. This beginner level eBook only skims the surface to expose you to some of the concepts and options to hand.

To cover the entire set of capabilities Power BI offers would take an encyclopaedia. However, **don't be put off**. You can achieve some very cool outcomes with very little learning needed.

This is where we hope this eBook can inspire you and help you with some of those first steps and concepts.

In doing so, we hope to give you a broad appreciation for the topic. One which also complements and may benefit from taking either our [Power BI JumpStart challenge](#) (takes 90 mins or less) and/or our [Power BI StepUp Challenge](#).

These items would form a great foundation to get you started into the world of Power BI.

What IS Power BI

Power BI consists of three main products:

Power BI Desktop

A Microsoft Windows desktop application. Download it here

<https://powerbi.microsoft.com/en-us/downloads/>

Power BI service

An online service (or SaaS – Software as a Service) which you use in the browser.

Power BI Mobile

Smartphone app (download it from the above link).

This eBook is only covering an introduction to the Power BI service (SaaS) and some aspects of the Power BI Desktop. To try these both out for yourself, try the [Power BI JumpStart challenge](#).

Furthermore, the data we are connecting to is an Excel file, a SharePoint list, a Microsoft Form, and data pulled from the web, particularly a Wikipedia entry.

The Difference between Power BI Desktop and Power BI service

Power BI Desktop is – put simply - more powerful as a ‘build’ tool than the Power BI service. The Service is more capable as a Sharing tool.

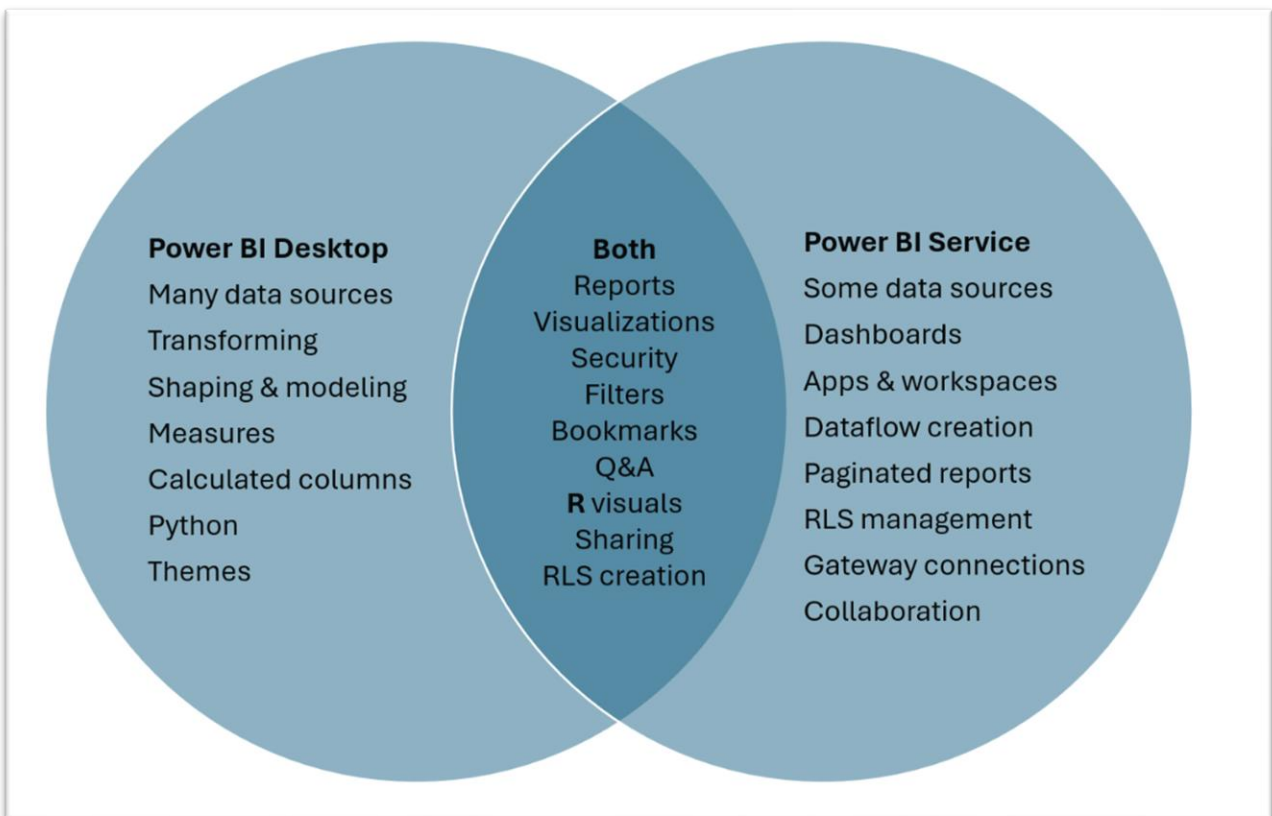
So, if you just need to generate a report from some pre-existing data already in the cloud service, you can use the Power BI service.

If you are Report and Dashboard builder, creating a combined set of data tables from multiple sources (called a **Data Model**) you will want to reach for the Desktop version first before publishing to the Power BI service or a mobile app.

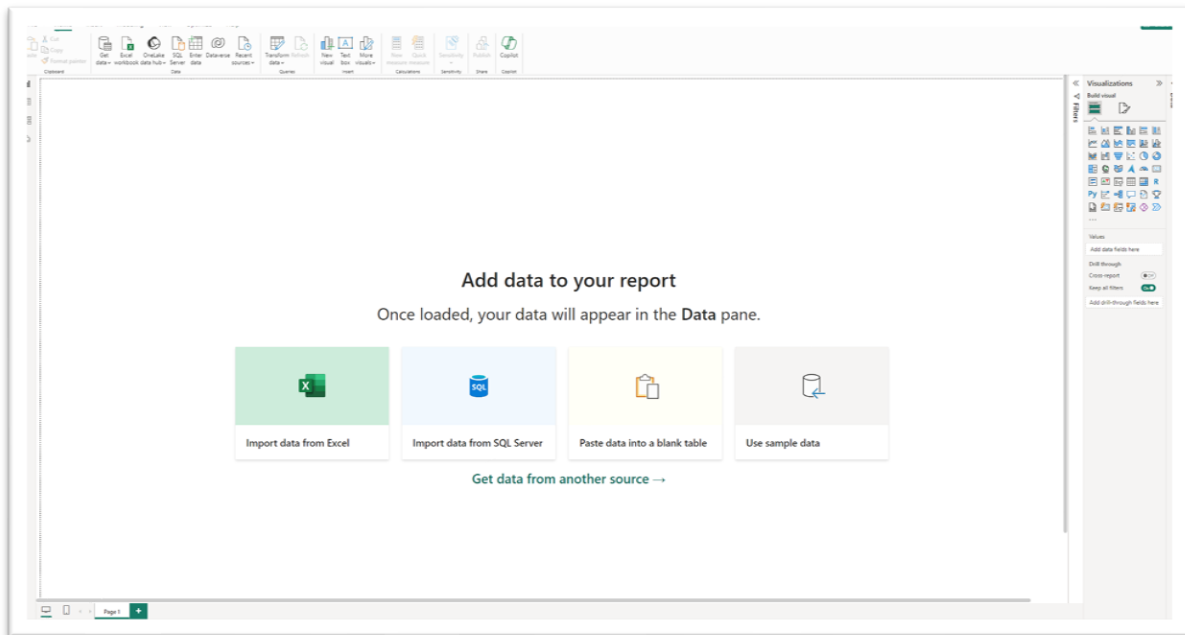
Build mostly in the Desktop, publish to the cloud service and share from there is how we think of it.

The below Venn diagram is a neat summary of the major differences of the 2 products.

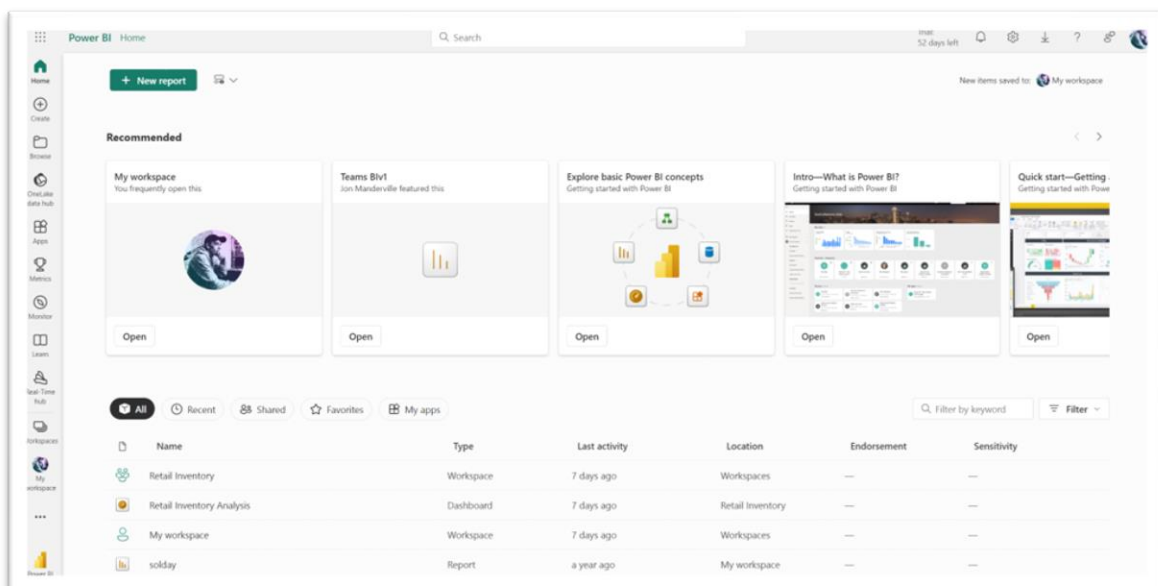
For more information, see <https://learn.microsoft.com/en-us/power-bi/fundamentals/service-service-vs-desktop>



Below is Power BI Desktop.



Below is the home of the Power BI service, which sits in a browser. As this is a browser-based product, it is connected to Microsoft 365 - hence the app launcher (9 dots) on the top left-hand side.



Power BI and Mobile Devices

Power BI has downloadable mobile apps which allow you to view and interact with content you have created or have access to on your smartphone and other devices.

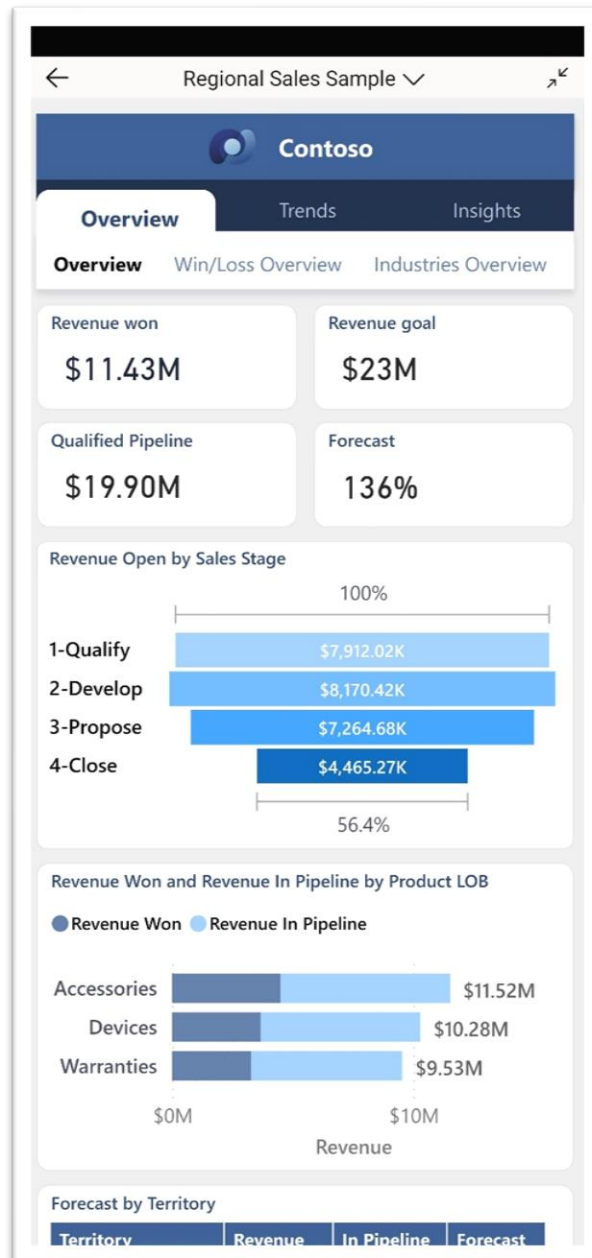
You create reports for mobile Power BI apps using the Power BI Desktop. It has an inbuilt designer specifically for mobile reports that you want to publish.

Device	Highlights
iPhone	iPhones go everywhere, and the Power BI mobile app for the iPhone goes, too. Besides viewing your Power BI reports in a special mobile layout view, you can add Power BI to your Apple Watch, and ask questions with the Q&A virtual analyst. Get started with the Power BI mobile app for iOS.
iPad	On the iPad, the Power BI mobile app displays dashboards and reports the way they were formatted for the Power BI service. Plus you can view your Power BI Report Server and Reporting Services KPIs and reports right on your iPad. You can set data alerts in the Power BI mobile app to notify you when data in a dashboard changes beyond limits you set. Get started with the Power BI mobile app for iOS.

Android phone	The Power BI mobile app for the Android phone brings Power BI to your pocket, with up-to-date, touch-enabled mobile access to your business information. You can filter a report by your geographic location. You can scan QR codes with your Android phone and go straight to a Power BI dashboard or report. Get started with the Power BI mobile app for Android.
Android tablet	This mobile app runs on a number of different Android tablets, bringing you up-to-date, touch-enabled mobile access to your business information. On the Android tablet, the Power BI mobile app displays dashboards and reports the way they were formatted for the Power BI service. You can mark your favourite dashboards and reports, so you can get to them quickly, along with your favourite Power BI Report Server and Reporting Services KPIs and reports. Get started with the Power BI mobile app for Android.
Windows	The Power BI mobile app for Windows 10 runs on any Windows 10 device, including Windows 10 phones. Along with many of the features of the other mobile apps, the Power BI mobile app for Windows 10 offers special functionality. For example, you can pin a Power BI dashboard to the Windows Start screen from the Power BI mobile app. Plus you can run Power BI in presentation mode on Surface Hub and in the Power BI mobile app for Windows 10. Get started with the Power BI mobile app for Windows 10 devices.

Table source: <https://docs.microsoft.com/en-us/power-bi/consumer/mobile/mobile-apps-for-mobile-devices>

The image below is a screenshot of a Power BI Android app:



Key Power BI concepts

The following are some key concept or terms you will see a lot in relation to Power BI.

Term	Definition
Workspaces	Workspaces are ‘containers’ for dashboards, reports, semantic models, and dataflows. They allow organisation of these items along with collaboration. They are also the launchpad for sharing your content beyond just yourself.
Semantic Model	A Semantic Model is a collection of data that you have created, imported or connected to. Some people refer to this as a Data Model. Power BI lets you connect to and import all sorts of data sources and bring them all together into your Semantic Model. Common terms you will hear in relation to models are: • Sources – Data sources • Tables - sets of data e.g. customer information • Cardinality - the relationship between data in separate tables e.g. 1 to many which is 1 customer in a table to many orders in an order table) • Dimension tables – Describes the business entities or things you model e.g. products • Fact tables – store observations or events about those entities e.g. orders.

Reports	A Power BI report is one or more pages of visualizations such as line charts, maps, and treemaps. Visualizations are also called visuals. You can: • Create reports from scratch within Power BI • import them from dashboards that colleagues share with you • Power BI can create them for you when you connect to data from Excel, Power BI Desktop, databases, and SaaS applications.
Table source: https://docs.microsoft.com/en-us/learn/modules/introductionhttps://docs.microsoft.com/en-us/learn/modules/introduction-power-bi/2-what-power-bipower-bi/2-what-power-bi	

Bringing data To Power BI

When you use Power BI. Typically, you will follow an ETL process. This means:

- **Extract** – gather data from the locations you need by connecting to it
- **Transform** – ensure the information is well formatted, free of errors and has been prepared to make sure it makes sense and can be read by Power BI.
- **Load** – Data has been added to your **Data Model**

Once this is done, you can create visualisations and prepare them for Publishing.

Before embarking on the **Extract** process, here are a few tips to ensure that your data is suitably prepared.

While you don't have to be a data scientist to use Power BI, the quality of the results and trends you might observe from your data will only be as good as the data itself.

The tips below for the preparation of an Excel file but may be used for other data sources, too.

While Power BI allows you to clean up your data once it has been imported during the **Transform step** (and this is necessary if you are bringing in data from, for example, the web and you have no control over the creation of the data), it's a good idea to try to have the cleanest data as possible from the start.

- All the data in each source column should be of the same type of data, for example, it must be a number, or text, date or a currency. A source tool like Excel will help you do this but if you don't have control, don't worry too much. You'll check during the **Transform** step.
- Think hard about what the purpose of your data is.
 - Do you need all the information?
 - What columns can you remove?
 - Perhaps you want to add a column from elsewhere.
- In short, reduce your data to its absolute minimum before you use it. Note: if you connect your Excel file to Power BI via OneDrive, if you update the OneDrive file, it will be updated in Power BI
- Where you have pivot tables or matrix style tables, put your data into a list with repeating values - rather than the pivot table for example where it is cross referenced
- Don't have any **totals** rows in your data source
- If your data is not actually a 'table' turn it into a table by selecting Format as a table in Excel. This will put a header above each column which will, in turn, be understood as the column name in Power BI when you perform the load step
- Within Excel, name your table and leave out spaces in the table name

Below is a good example of what is called “clean” data copied from an Excel file.

Each column is its own data type with simple, functional columns.

You can copy and paste it into Excel to follow our example. Save it somewhere you can access easily.

OrderDate	Region	Rep	Item	Units	Unit Cost		Total	
1/6/19	East	Jones	Pencil	95	£	1.99	£	189.05
1/23/19	Central	Kivell	Binder	50	£	19.99	£	999.50
2/9/19	Central	Jardine	Pencil	36	£	4.99	£	179.64
2/26/19	Central	Gill	Pen	27	£	19.99	£	539.73
3/15/19	West	Sorvino	Pencil	56	£	2.99	£	167.44
4/1/19	East	Jones	Binder	60	£	4.99	£	299.40
4/18/19	Central	Andrews	Pencil	75	£	1.99	£	149.25
5/5/19	Central	Jardine	Pencil	90	£	4.99	£	449.10
5/22/19	West	Thompson	Pencil	32	£	1.99	£	63.68
6/8/19	East	Jones	Binder	60	£	8.99	£	539.40
6/25/19	Central	Morgan	Pencil	90	£	4.99	£	449.10

7/12/19	East	Howard	Binder	29	£	1.99	£ 57.71
7/29/19	East	Parent	Binder	81	£	19.99	£ 1,619.19
8/15/19	East	Jones	Pencil	35	£	4.99	£ 174.65
9/1/19	Central	Smith	Desk	2	£	125.00	£ 250.00
9/18/19	East	Jones	Pen Set	16	£	15.99	£ 255.84
10/5/19	Central	Morgan	Binder	28	£	8.99	£ 251.72
10/22/19	East	Jones	Pen	64	£	8.99	£ 575.36
11/8/19	East	Parent	Pen	15	£	19.99	£ 299.85
11/25/19	Central	Kivell	Pen Set	96	£	4.99	£ 479.04
12/12/19	Central	Smith	Pencil	67	£	1.29	£ 86.43
12/29/19	East	Parent	Pen Set	74	£	15.99	£ 1,183.26
1/15/20	Central	Gill	Binder	46	£	8.99	£ 413.54
2/1/20	Central	Smith	Binder	87	£	15.00	£ 1,305.00
2/18/20	East	Jones	Binder	4	£	4.99	£ 19.96
3/7/20	West	Sorvino	Binder	7	£	19.99	£ 139.93
3/24/20	Central	Jardine	Pen Set	50	£	4.99	£ 249.50

4/10/20	Central	Andrews	Pencil	66	£	1.99	£	131.34
4/27/20	East	Howard	Pen	96	£	4.99	£	479.04
5/14/20	Central	Gill	Pencil	53	£	1.29	£	68.37
5/31/20	Central	Gill	Binder	80	£	8.99	£	719.20
6/17/20	Central	Kivell	Desk	5	£	125.00	£	625.00
7/4/20	East	Jones	Pen Set	62	£	4.99	£	309.38
7/21/20	Central	Morgan	Pen Set	55	£	12.49	£	686.95
8/7/20	Central	Kivell	Pen Set	42	£	23.95	£	1,005.90
8/24/20	West	Sorvino	Desk	3	£	275.00	£	825.00
9/10/20	Central	Gill	Pencil	7	£	1.29	£	9.03
9/27/20	West	Sorvino	Pen	76	£	1.99	£	151.24
10/14/20	West	Thompson	Binder	57	£	19.99	£	1,139.43
10/31/20	Central	Andrews	Pencil	14	£	1.29	£	18.06

Creating a Workspace and Report in Power BI

When you think about a Power BI workspace, think of a container which houses dashboards, reports, datasets, and dataflows. There are two types of workspaces:

- My workspace
- Workspaces.

My workspace is a personal workspace where you can work on your own content whereas **Workspaces** can be collaborative spaces which are used to work on Power BI content with other people ideally.

Getting data from Excel

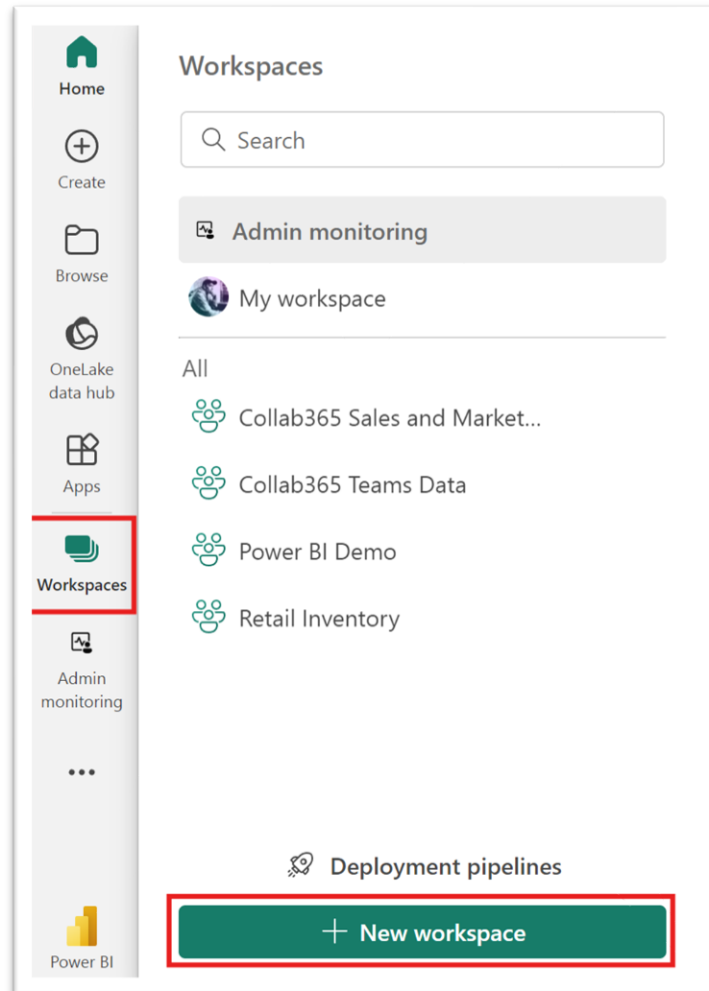
For our first Workspace and report, we are going to use the above data which has been copied to create an Excel file – you can store locally wherever is convenient.

Go ahead and make this file now

We're going to upload it to the Power BI service and present the data in a visually appealing way. Furthermore, we'll export the data to Teams and SharePoint online.

Head to app.powerbi.com and sign in to your Microsoft 365 account. You may need to create a trial to follow these steps (generally you will see on screen prompts inviting you to do this if you are not licenced already).

Within the Power BI Service, Select **Workspaces** → Create a workspace.



You have the option to upload an image, give it a name and a description. We're going to upload an image of stationary and call it EbookWorkspace.

(Note: with the image below, from time-to-time Power BI screens will change, new features are added and older ones removed.

You may need to use a little lateral thought over time if something does not quite look the same, however at the time of writing, Workspace menu remains very similar to the image below).

Create a workspace

Name *

Name this workspace

Description

Describe this workspace

Domain ⓘ

Assign to a domain (optional)

Learn more about workspace settings

Workspace image

Upload

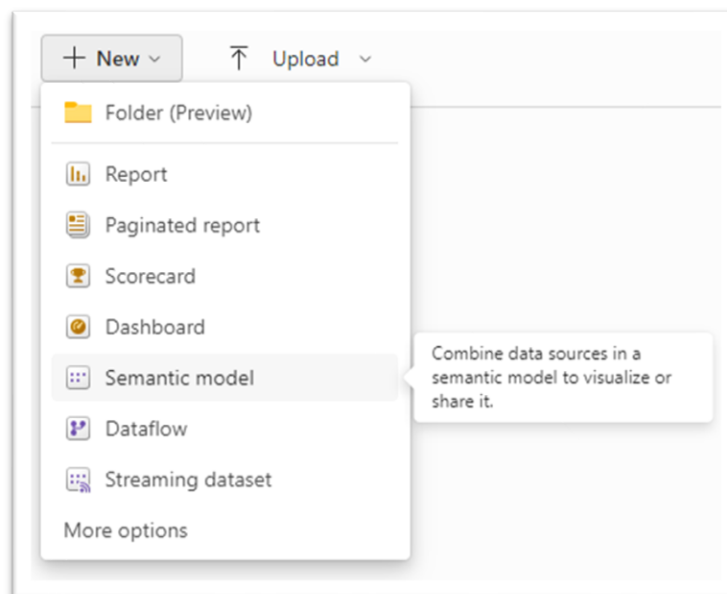
Reset

Advanced

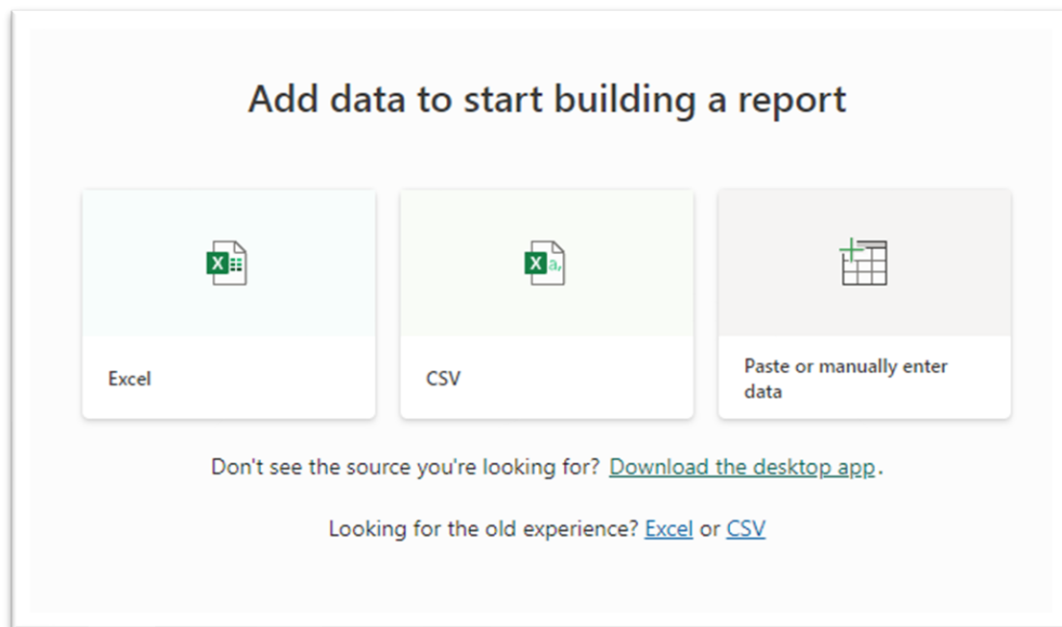
Apply

Cancel

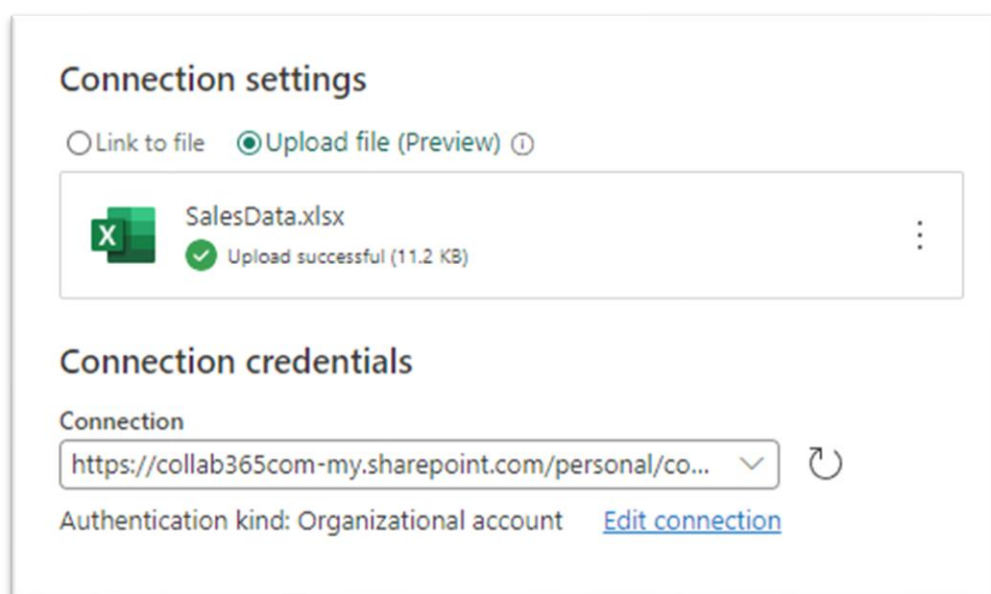
When our workspace is created, it will be empty. We're going to select **New** and **Semantic Model** to Create a data to use in a report.



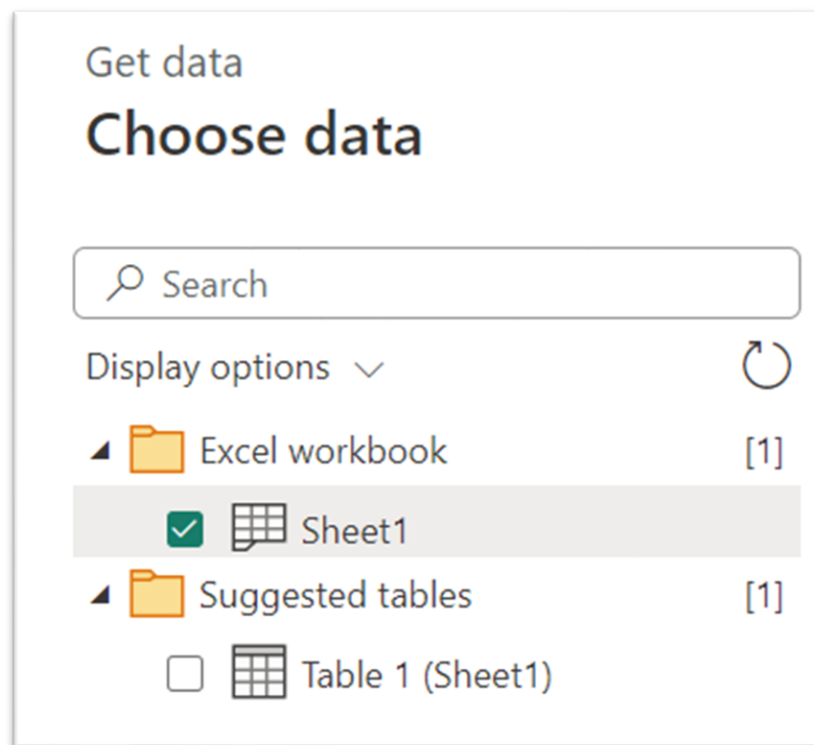
Click **Excel** to import your data from an excel workbook.



Select **Upload file** and choose your Excel file we made earlier (we called ours **Sales Data**). You may need to sign in to your Microsoft account at this stage.

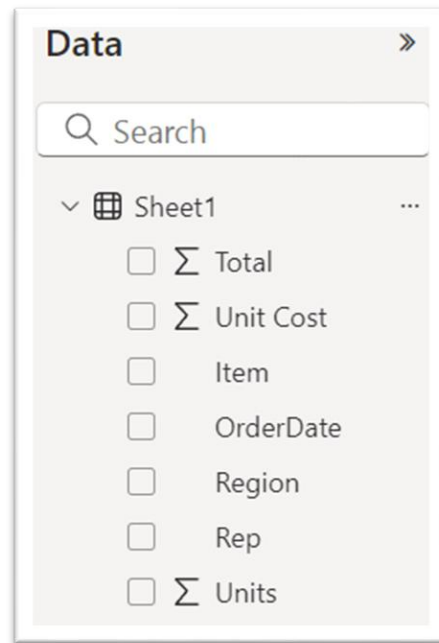


You'll now see the Power Query interface, where you'll find a list of sheets and tables that exist in your Excel workbook. (If no tables exist – it will automatically try and create one for you). For this eBook, we are going to select the **Sheet1** from the **Excel workbook** folder and click **Create**.



After a few moments your data will be processed and you'll see a new interface where we can create visualizations.

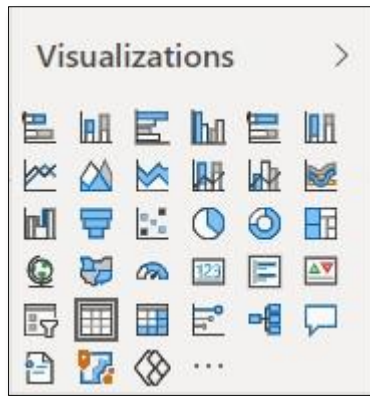
First, let's look at **Fields** which is on the right hand-side of the screen (You may need to expand the data to see the fields). Here you will see the data as referenced in the original Excel file:



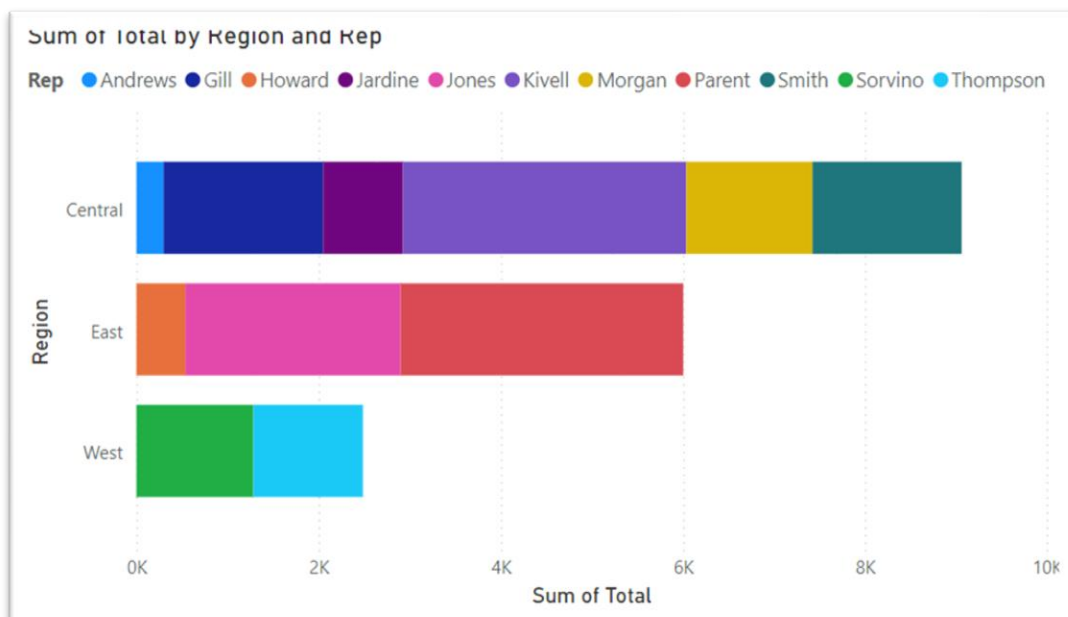
If we select (tick) *Region*, *Rep* and *Total*, we see the raw data in a table within the canvas area:

Region	Rep	Sum of Total
Central	Andrews	298.65
Central	Gill	1,749.87
Central	Jardine	878.24
Central	Kivell	3,109.44
Central	Morgan	1,387.77
Central	Smith	1,641.43
East	Howard	536.75
East	Jones	2,363.04
East	Parent	3,102.30
West	Sorvino	1,283.61
West	Thompson	1,283.61
Total		17,554.21

However, we can make this more interesting by using the many options that appear in Visualizations. With the table selected, try clicking another visualisation and watch the table change. (Tip: Hover over the visualisation to see its name)

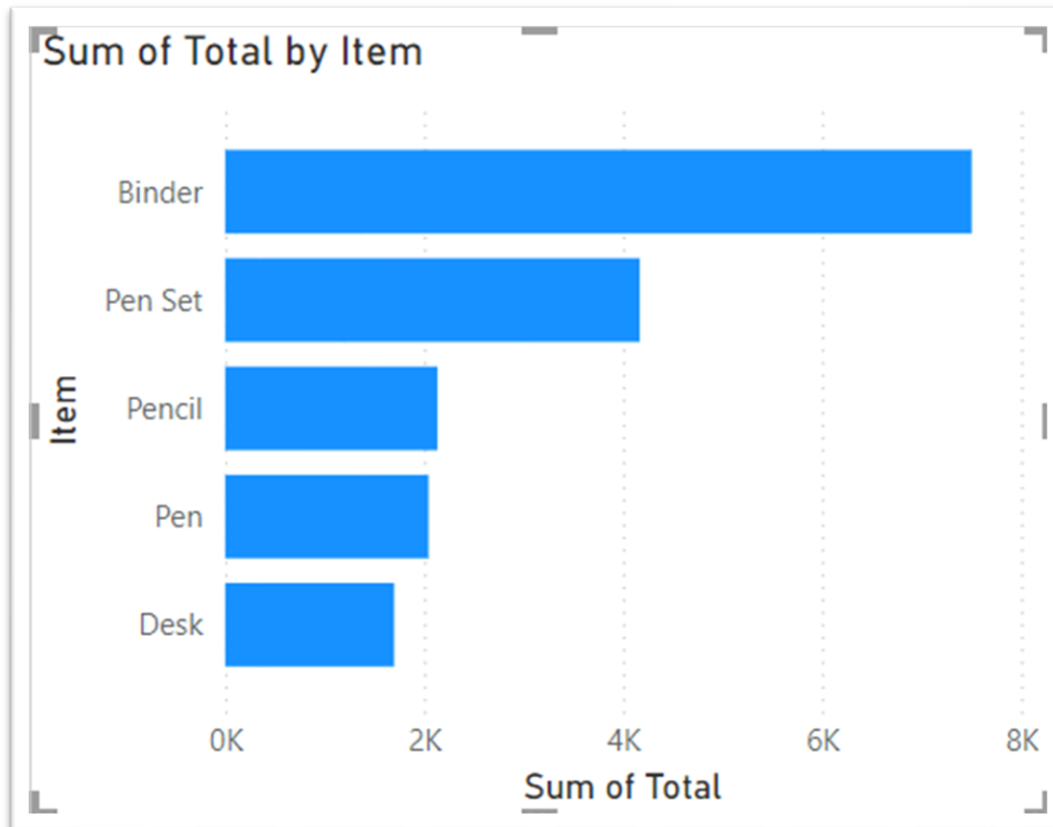


Below we have used a **Stacked bar chart** and have enlarged the chart in order to see all the *reps*. To make any chart bigger, just stretch it from the bottom corner.



It's always wise to save your report by selecting the **Save icon** (top right) or using **File -> Save**.

The next visualization we'll do is a **Clustered bar chart** and we're selecting the **Item** and **Total** (You'll need to ensure the previous visualization is unselected before creating a new visualization).



Once you save your report, you can perform various actions such as export it, share it with others or even publish to what's called an App. An App is essentially a browser version of your report ready for others to interact with. Each workspace may have up to 1 App.

We won't create an App yet but it's worth being aware that it is a step you may take when you publish.

Sharing data from Power BI

Now that we have created our first report based on our Excel file, we can export it or share it. It can be exported to PowerPoint, as PDF or back to Excel.

From your workspace, you can click the **+New > report** and see it in your browser.

Notice there is a **Share** button at the top.

Under the options for **Share**, there are the familiar sharing options from other Microsoft 365 products such as OneDrive and SharePoint.

However, if you click into the **Options**, the settings include the following:

➔ Allow recipients to build content with the data associated with this report (by default it is deselected).

Please note the following conditions when it comes to sharing:

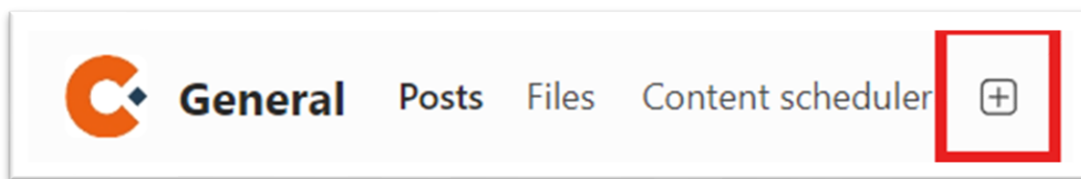
- You can share with people who are inside and outside of your organisation
- When you share a report, generally they can view it, they can hover over the visualisations to glean more data, but they cannot edit it
- You can allow people in your organisation to share with other people in your organisation
- You can share from Favourites, Recent, My Workspace and from other workspaces if you have the permission
- **When you share you must have a Power BI Pro or Premium Per User license.** Also, the people that you share with also need the same license. To see your report, they must sign into Power BI. External people can't edit content

For more about sharing, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/collaborate-share/service-share-dashboardsbi/collaborate-share/service-share-dashboards>

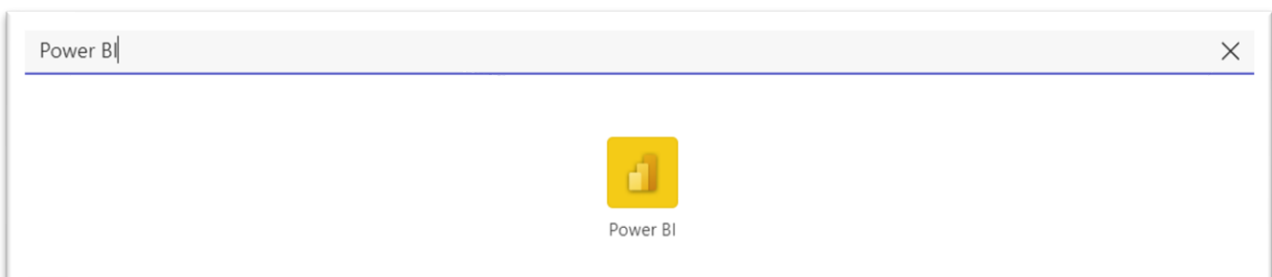
Sharing data in Teams

You can share your report to your colleagues within Teams & Channels.

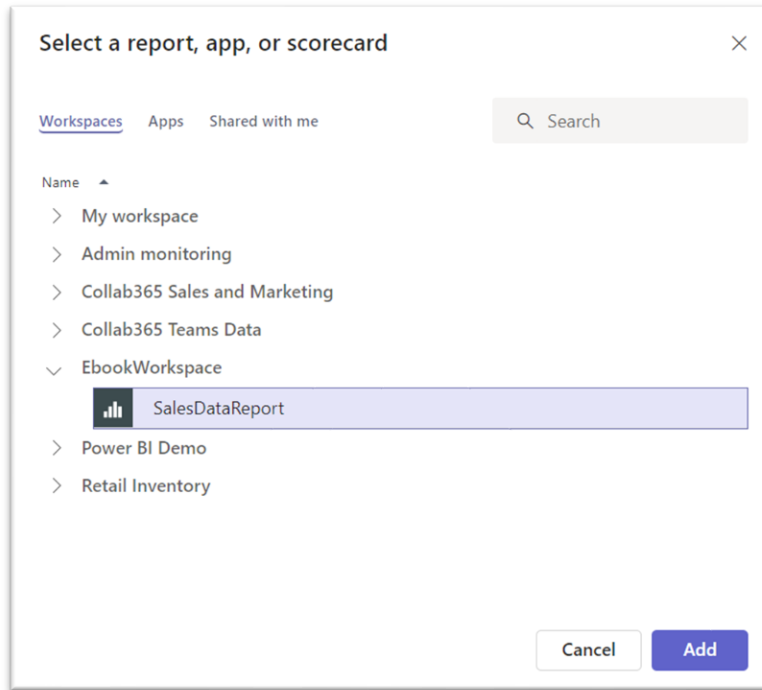
Our suggested approach would be to navigate to the Teams Channel you would like to add your report to and add a new tab by selecting the + icon at the top of the teams interface.



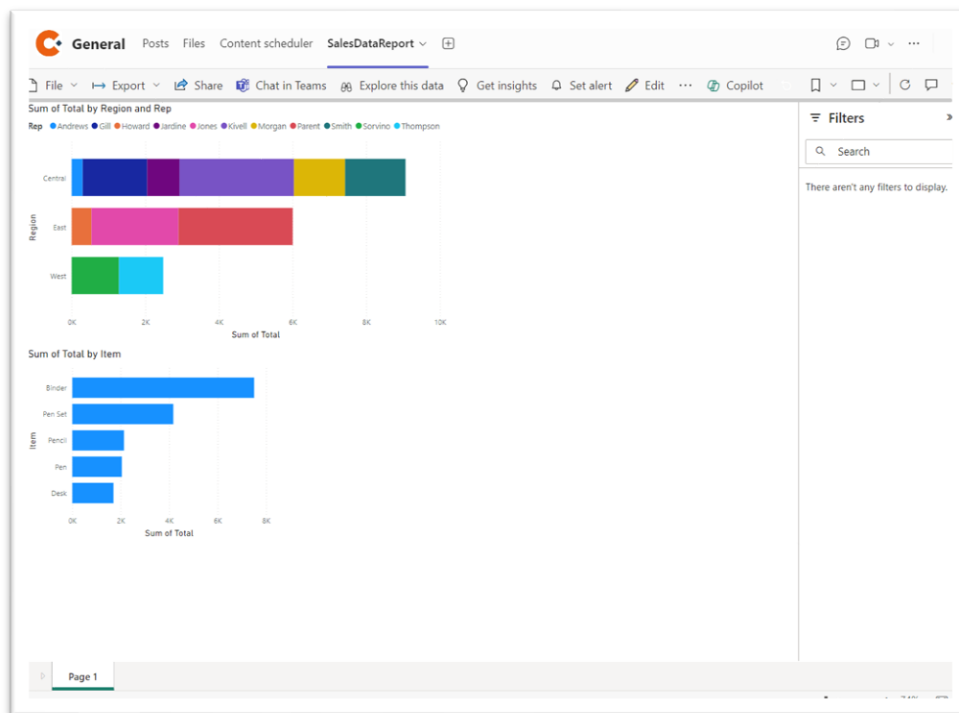
Search for and select the Power BI app from the interface. (Once selected you will need to save your tab first before being able to configure it with a report). You can also rename your tab by selecting the small drop-down arrow next to it and clicking **Rename**.



You'll now be able to paste a Power BI link into the input box or navigate to it by **Browse workspaces**, finding the workspace that your report was published to and selecting it from the list, the click **Add**.

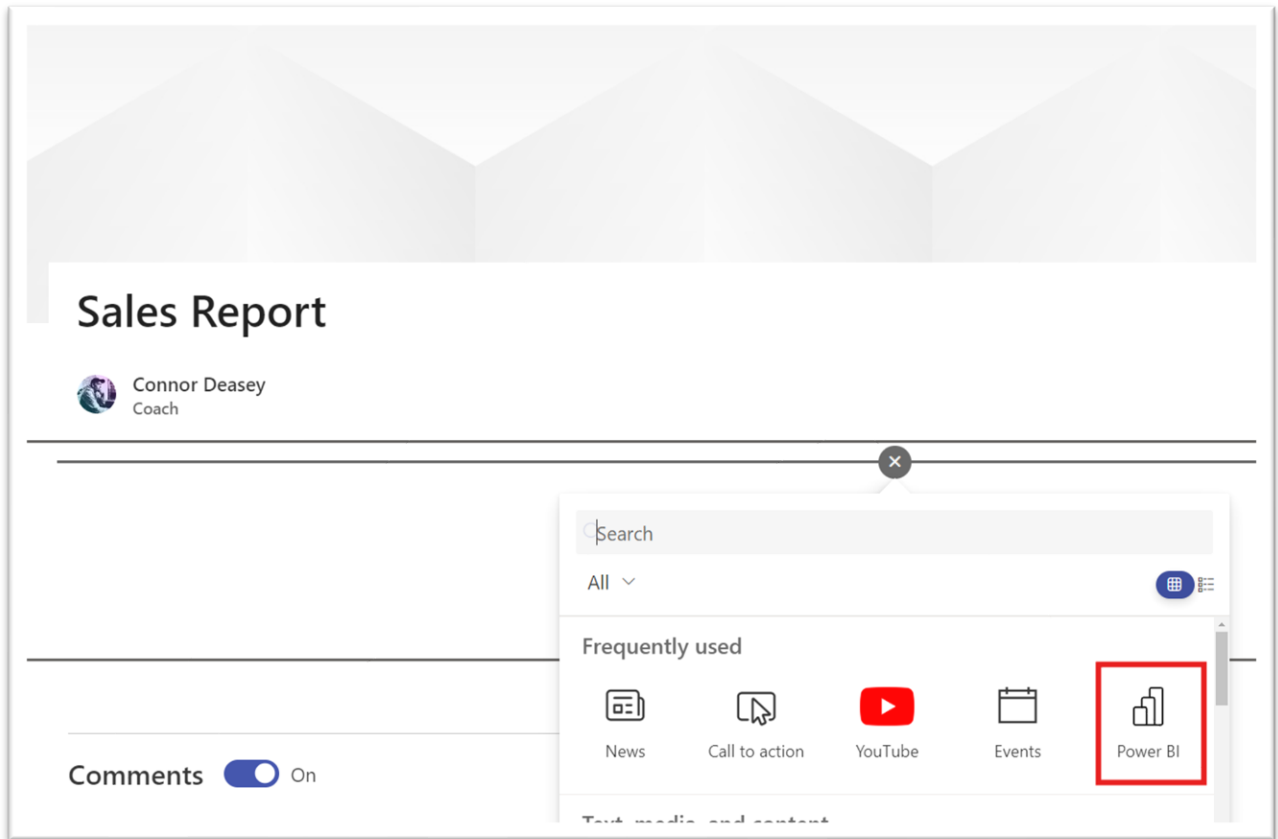


It appears in the Teams channel as a tab and can be shared and viewed by people in your organization

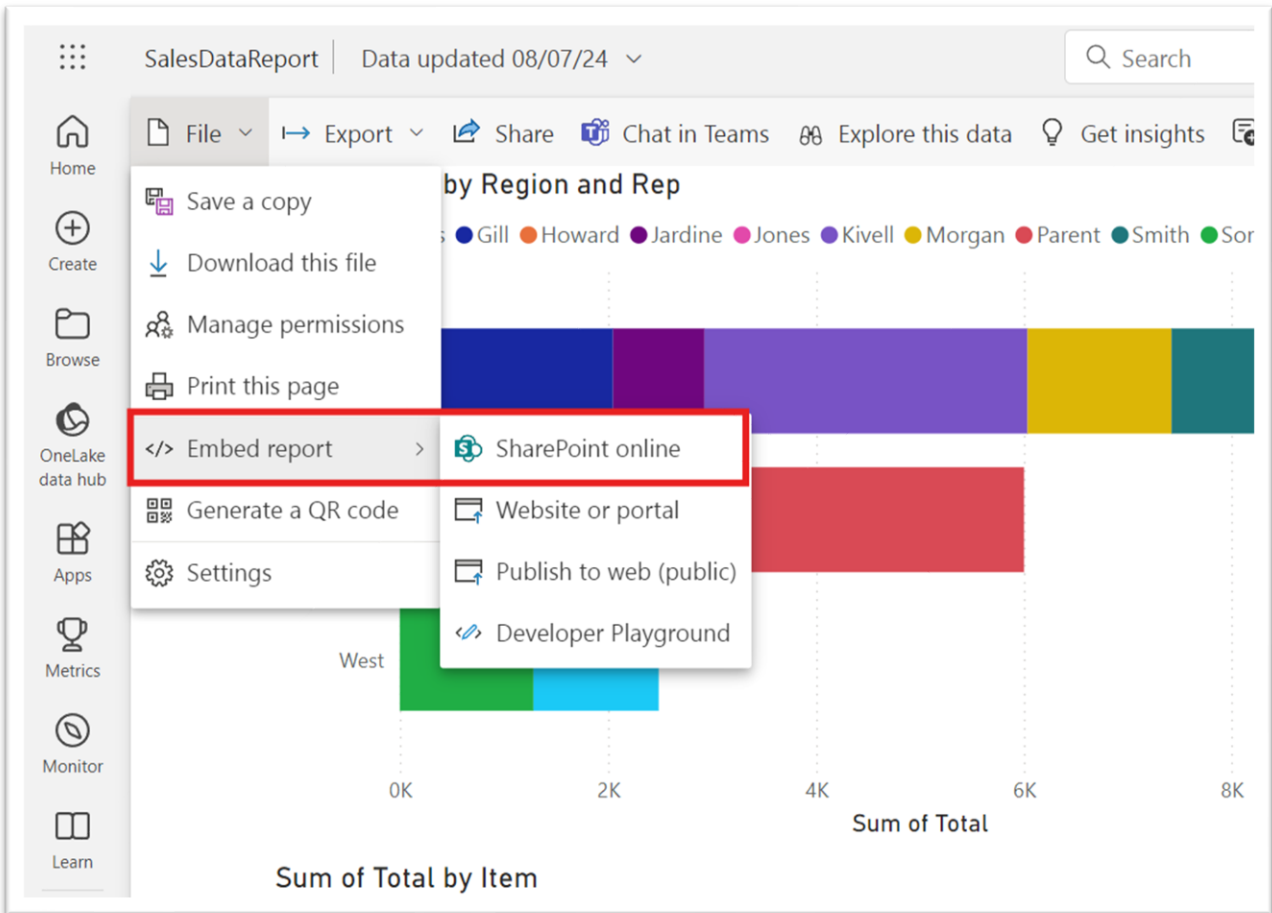


Sharing data in SharePoint

Go to the SharePoint site where you want to publish the report. Create a page and then add the Power BI webpart:



You will be prompted to add a report by inserting a link. You can retrieve this back in Power BI by selecting **File -> Embed report -> SharePoint Online** on the report you want to embed.



Note

- You need a Power BI Pro or Premium Per User (PPU) license to create a report in a workspace.
- To share with *Microsoft free users*, the workspace needs to be in a *Premium capacity*.

For more information, see here <https://docs.microsoft.com/en-us/power-bi/collaboratehttps://docs.microsoft.com/en-us/power-bi/collaborate-share/service-embed-report-spshare/service-embed-report-spo>

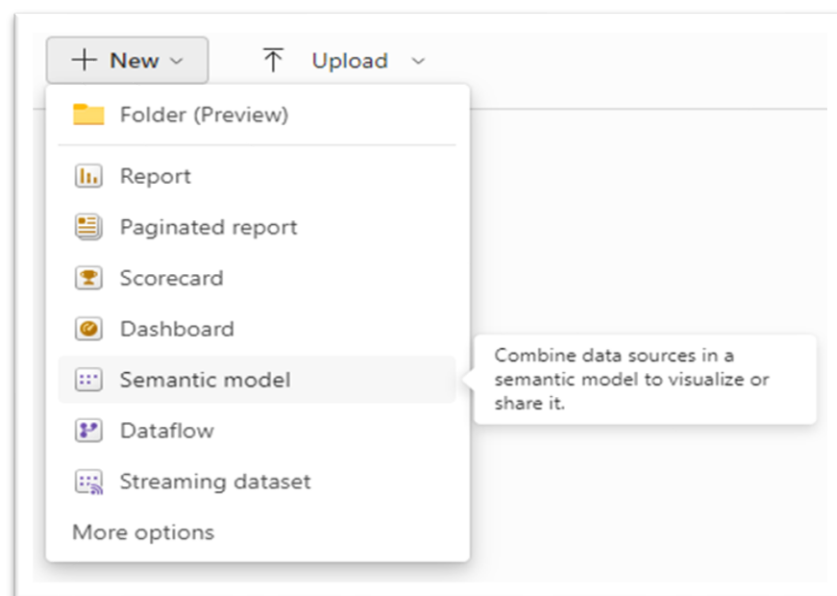
Getting data from OneDrive for Business

For this example, we've downloaded a sample file from Microsoft

(<https://docs.microsoft.com/en-us/power-bi/create-reports/sample-financial-download>)

which has a lot of dummy data. It's a great file to use as it's from a trusted source and has lots of data in it. You'll need to save this to a location in your OneDrive account.

On our EbookWorkspace workspace, we're uploading it by selecting **New -> Semantic Model** and select **Excel** in the center of your screen.



This time we're going to select a file which sits in your OneDrive, or you have the permission to access in another user's OneDrive. This has the advantage over uploaded from your PC as Power BI will check the file periodically to see if there are any changes; and if there are, the changes will be updated to Power BI.

Connection settings

☒ Link to file ☐ Upload file (Preview) ⓘ

File path or URL *

Select the **Link to file** option, then select the **Browse OneDrive** button where you'll be granted with a familiar OneDrive interface where you can choose files from your personal or SharePoint OneDrive locations.

Choose the **Financial Sample** and **Select the Next**.

Browse OneDrive

Pick a file

My files

New folder

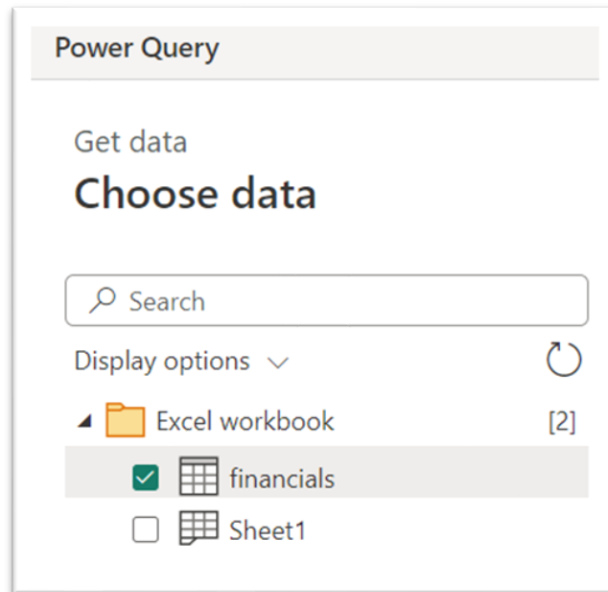
Apps	August 26, 2022	Connor Deasey	2 items	Private
Recordings	March 15, 2022	Connor Deasey	56 items	Private
Microsoft Teams Data	February 16, 2022	Connor Deasey	1 item	Private
BackUp 1901	January 18, 2022	Connor Deasey	22 items	Private
Notebooks	September 23, 2021	Connor Deasey	1 item	Private
BackUp	August 26, 2021	Connor Deasey	16 items	Private
Branding	July 6, 2021	Connor Deasey	0 items	Owner: Content Library
Microsoft Teams Chat Files	July 6, 2021	Connor Deasey	568 items	Private
Financial Sample.xlsx	32 minutes ago	Connor Deasey	81.7 KB	Private

1 selected

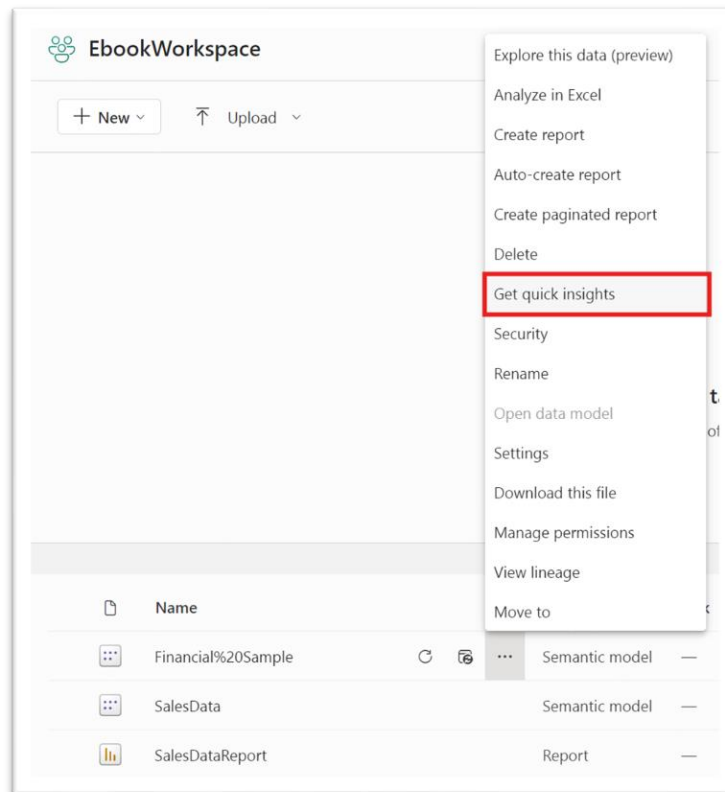
Select

Cancel

Just as we've done before, you'll need to select the Workbook from the Power Query screen. In this case, select the **financials** workbook and click **Create**.



Once it's in our workspace, we can return to our Power BI workspace. Find the **ellipsis** next to our **Financial Sample** Semantic Model and select **Get Quick Insights**.

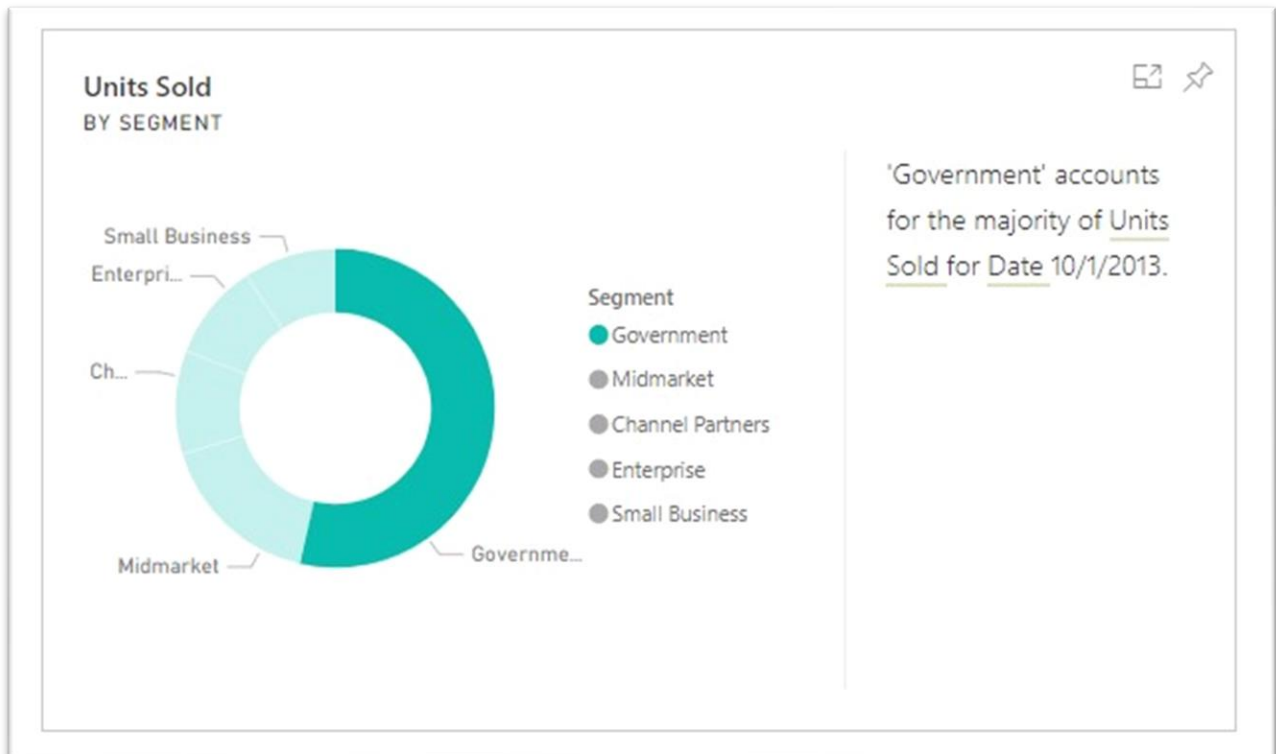
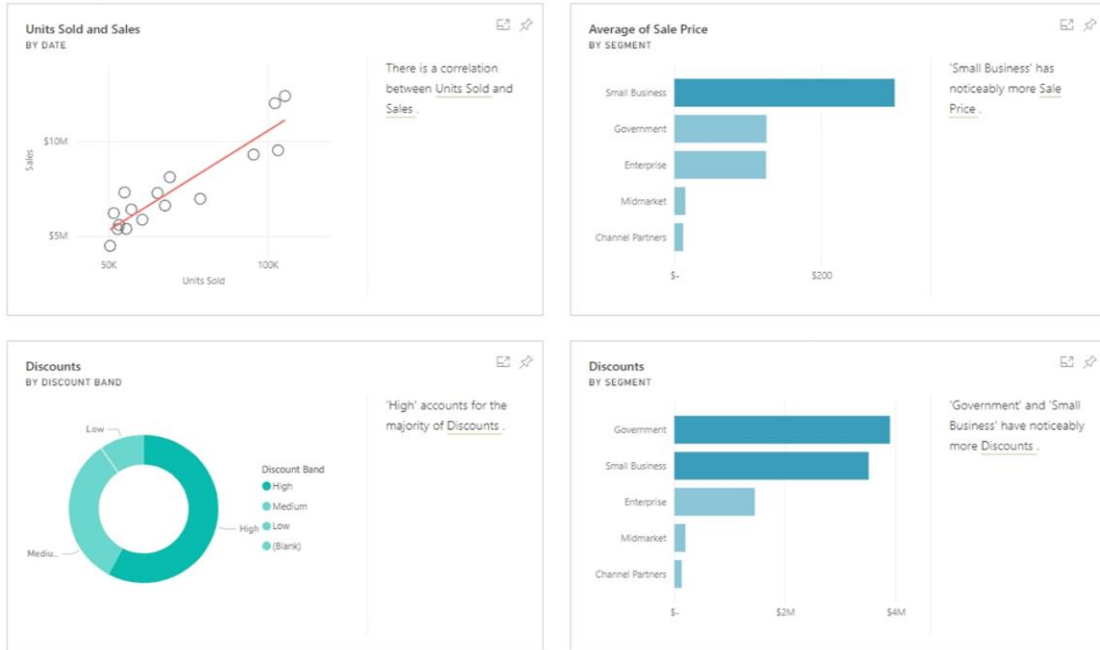


In the top right corner your insights will be generated and after a short period you'll be able to **View insights**.

Quick Insights will bring back up to 40 different visualizations structure in a way that Power BI deems best fits the data provided to it. Beneficial for those looking for "Quick Insights" into their data.

Quick Insights for Financial Sample

A subset of your data was analyzed and the following insights were found. [Learn more](#)

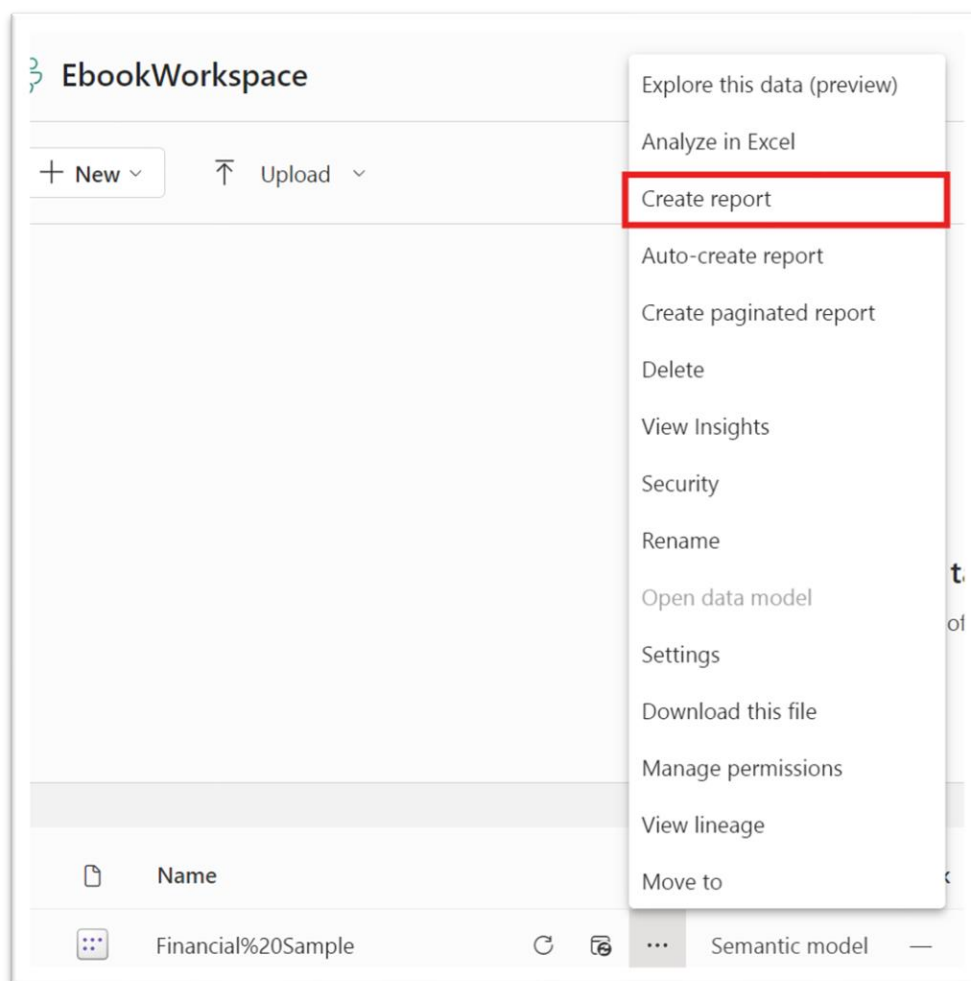


If you click on any of the visualizations, it will be enlarged thus giving a more detailed and interactive view.

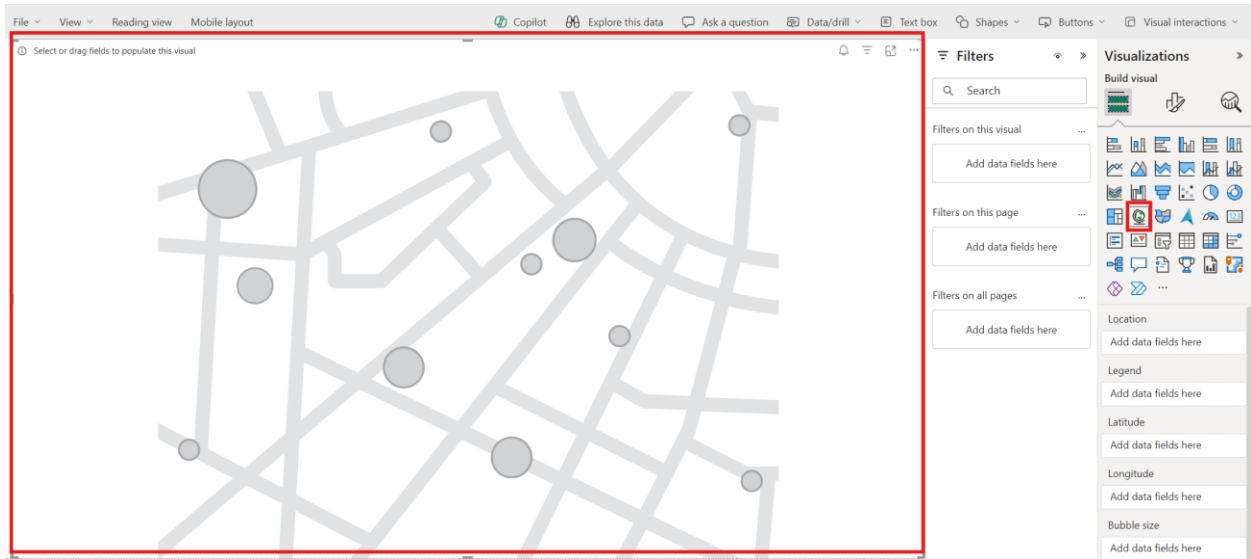
Turning your data into a map

In the next example, we're going to create a map to detail our sales on a world map.

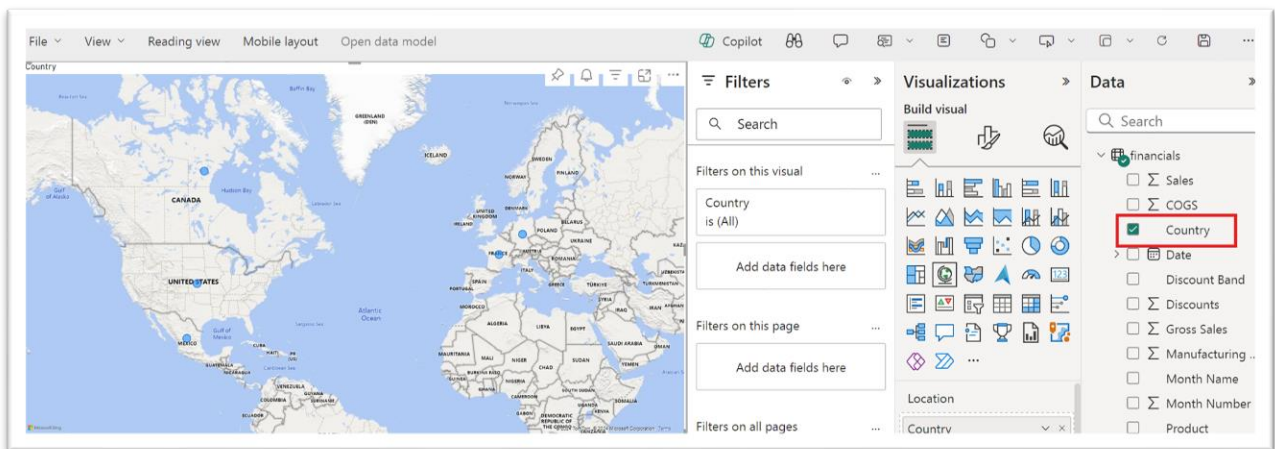
Find your **Financial Sample** Semantic Model in your workspace, select the **ellipsis** and click **Create report**.



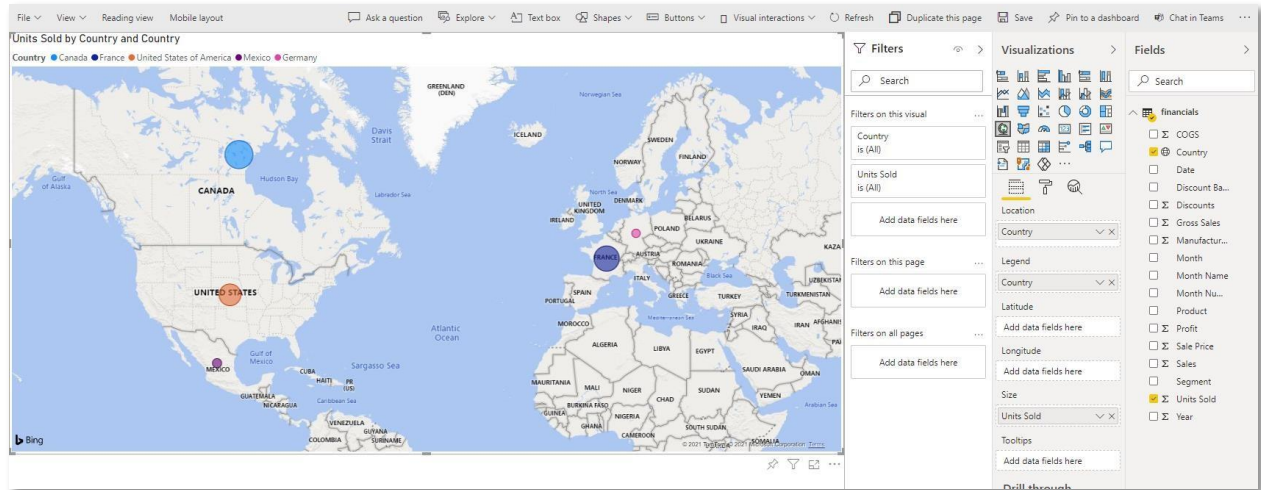
Select **Map** from the list of visualizations and using the handles, scale it to fit your report boundaries.



As you can see, it's still a blank map as have yet to select the Country field. Once we do, our map comes alive, and we see where our units are being proportionately sold; they appear as blue circles over Canada, US, Mexico, France, and Germany.



Next, we're going to add a legend which gives our countries a different colour.



Once you have it saved, it can be shared in the same way as our previous example i.e. to Teams, SharePoint Online, or exported as a PDF, Excel file or a PowerPoint presentation.

Using Power BI Desktop

Getting data from a SharePoint list

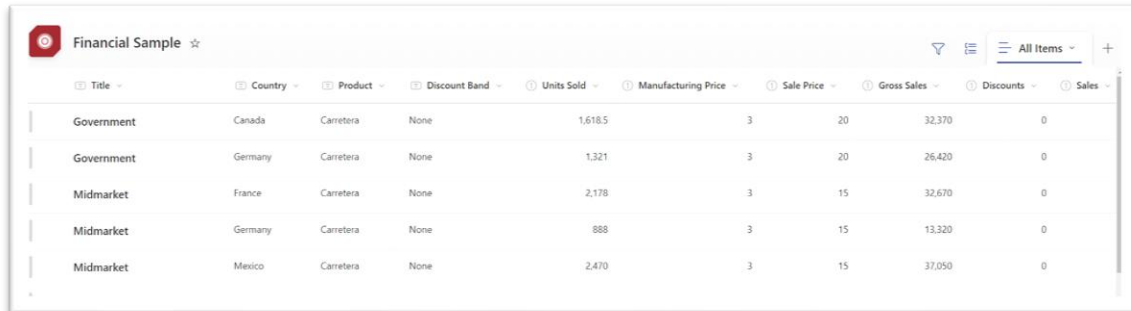
Although SharePoint lists are an ideal place to store data, within SharePoint itself it's not a particularly visual medium.

For example, below is a screenshot of a SharePoint list containing the financial data we downloaded earlier.

Before proceeding, import this data into a SharePoint list. There are plenty of tutorials on how to do this. This Academy session shows how to do this or do a quick search on google for "Import data Into A SharePoint Online list"

<https://members.collab365.com/c/beginners-library/an-introduction-to-sharepoint>

While the data is structured and easy to update, if we bring it into Power BI we can present it in a more appealing way.

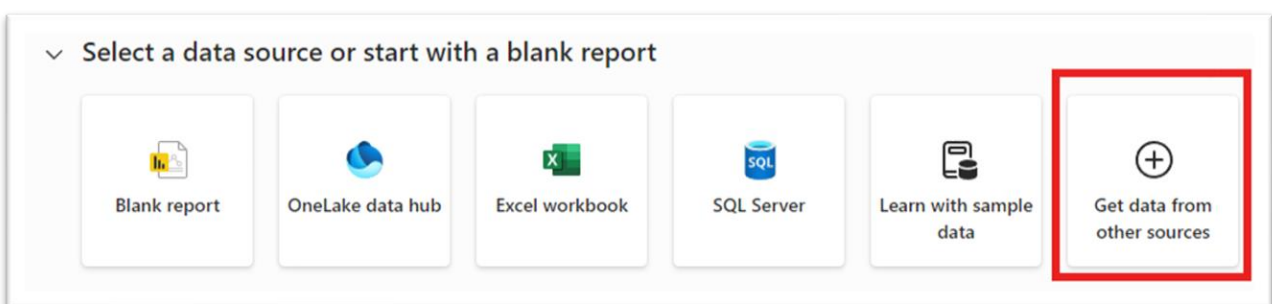


The screenshot shows a Power BI report titled "Financial Sample". It displays a table with the following columns: Title, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, and Sales. The table contains five rows of data.

Title	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales
Government	Canada	Carretera	None	1,618.5	3	20	32,370	0	
Government	Germany	Carretera	None	1,321	3	20	26,420	0	
Midmarket	France	Carretera	None	2,178	3	15	32,670	0	
Midmarket	Germany	Carretera	None	888	3	15	13,320	0	
Midmarket	Mexico	Carretera	None	2,470	3	15	37,050	0	

To help with this example, we are going to use **Power BI Desktop** for the first time. This opens new additional features for us that we may not have experienced in the Power BI Service.

Open Power BI Desktop and select **Get data from other sources**. Search for **SharePoint** and **connect** to a **SharePoint Online List** with your SharePoint Site URL (Use the Microsoft Account connection to authenticate).



Get Data

sharepoint ×

All

File

Online Services

Other

All

-  SharePoint folder
-  SharePoint Online List
-  SharePoint list

SharePoint Online Lists

Site URL ⓘ

https://collab365com.sharepoint.com/sites/ABCTeamSite/

Implementation

☐ 2.0

☒ 1.0

▸ Advanced options

OK

Cancel

Select your **Financial Sample** and click **Load**

Navigator

Display Options

<https://collab365com.sharepoint.com/sites/...>

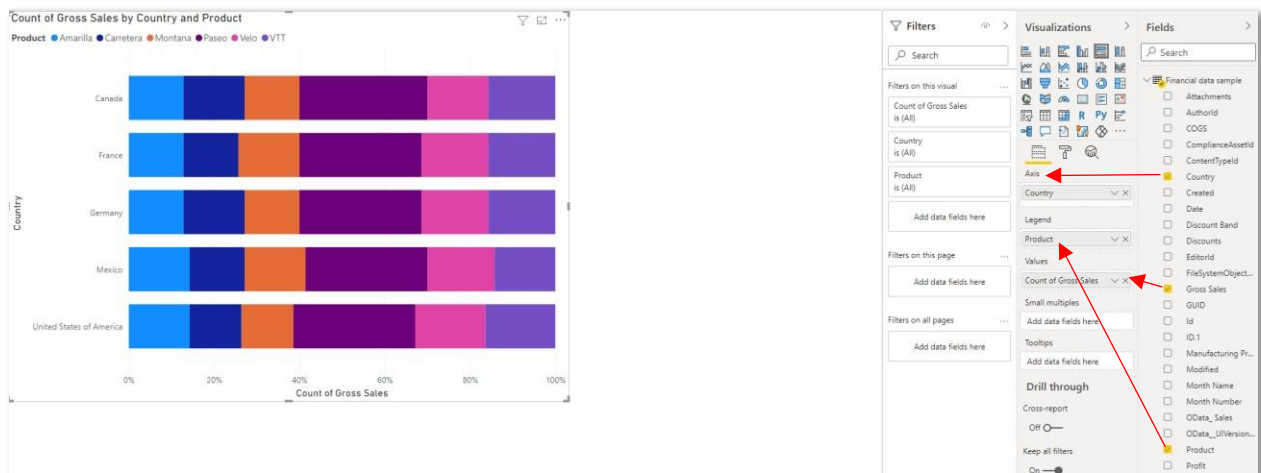
☐ appdata
☐ appfiles
☐ Composed Looks
☐ Converted Forms
☐ Documents
☐ Employee onboarding
☐ Events
☒ Financial Sample
☐ Form Templates
☐ List Template Gallery

Financial Sample

FileSystemObjectType	Id	ServerRedirectedEmbedUri	ServerRedirectedEmbed
	0	1	null
	0	2	null
	0	3	null
	0	4	null
	0	5	null
	0	6	null
	0	7	null
	0	8	null
	0	9	null
	0	10	null

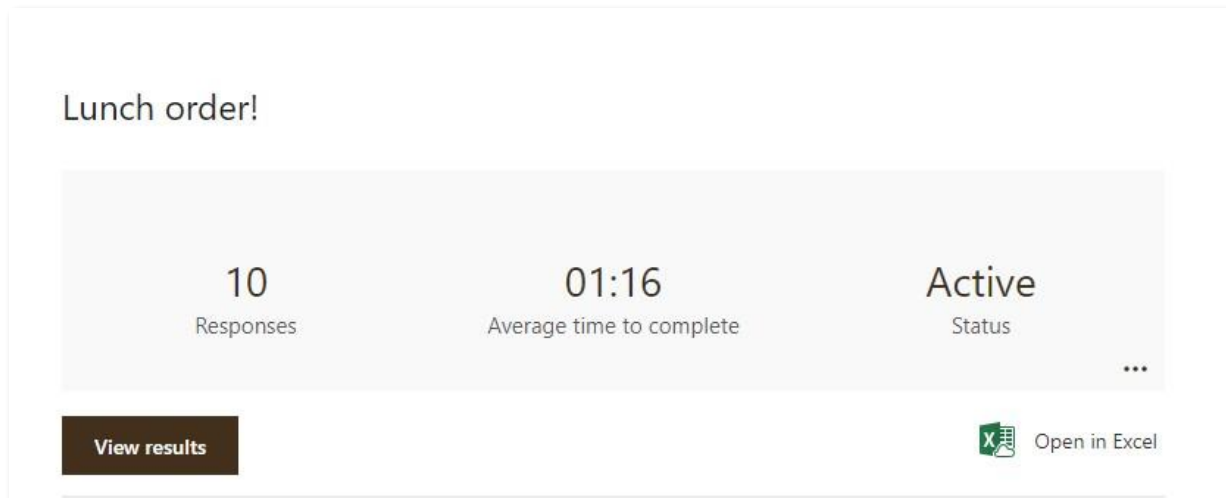
The data in the preview has been truncated due to size limits.

Select your visuals just as we have done before. The below example is a **Stacked bar chart** - a nice visual which details the Count of Gross Sales by Country and Product. (You may need to transform your data using the option under the Home tab. This is where you can rename columns and change the data type assigned to them)



Getting data from Microsoft Forms

Microsoft Forms is another product which collects exportable data. In the below example, we have a short survey of 10 responses.



To get the data into Power BI, we have two options. Export the data to Excel and then import it to Power BI or create a SharePoint list based on the Excel file, and then import or link it to Power BI.

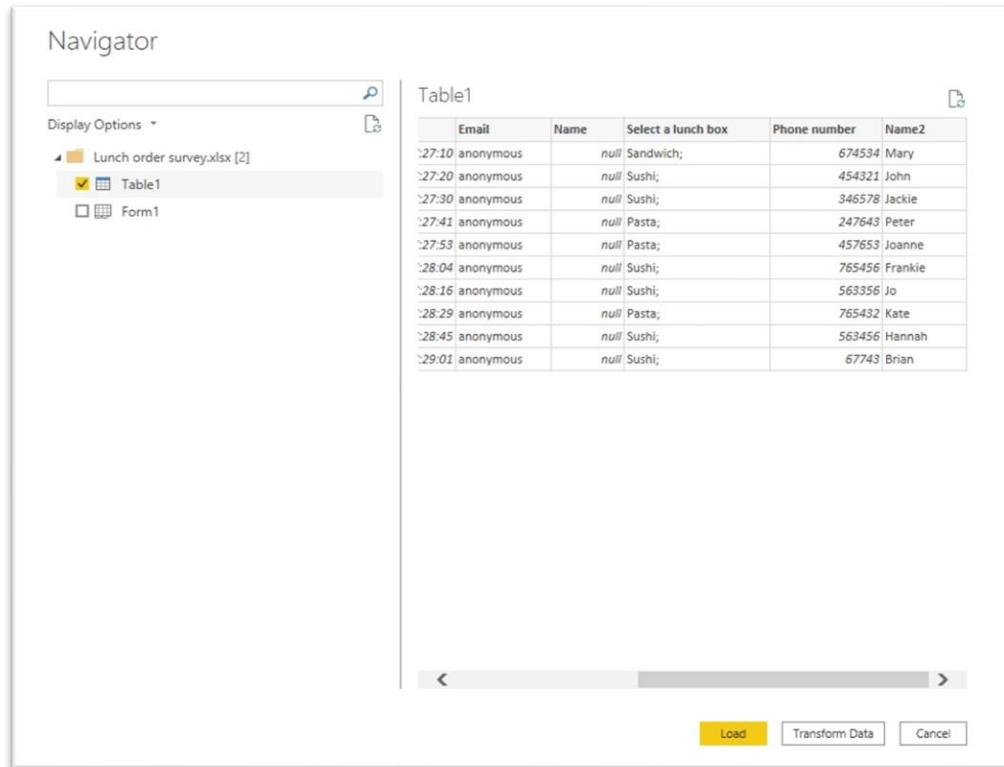
We're going to use option one, export to Excel.

Select **Open in Excel**. The data comes in structured in columns with column headers – this makes it easy to organise the data once in Power BI.

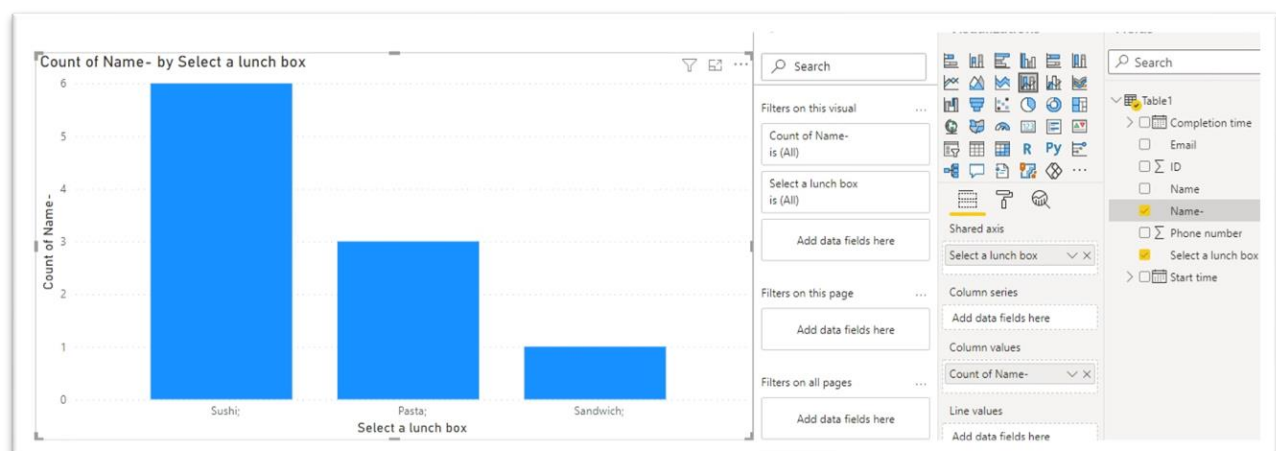
	B	C	D	E	F	G	H
	Start time	Completion time	Email	Name	Select a lunch box	Phone number	Name2
1	5/25/21 7:26:59	5/25/21 7:27:10	anonymous		Sandwich;	674534	Mary
2	5/25/21 7:26:56	5/25/21 7:27:20	anonymous		Sushi;	454321	John
3	5/25/21 7:26:53	5/25/21 7:27:30	anonymous		Sushi;	346578	Jackie
4	5/25/21 7:26:50	5/25/21 7:27:41	anonymous		Pasta;	247643	Peter
5	5/25/21 7:26:47	5/25/21 7:27:53	anonymous		Pasta;	457653	Joanne
6	5/25/21 7:26:44	5/25/21 7:28:04	anonymous		Sushi;	765456	Frankie
7	5/25/21 7:26:41	5/25/21 7:28:16	anonymous		Sushi;	563356	Jo
8	5/25/21 7:26:37	5/25/21 7:28:29	anonymous		Pasta;	765432	Kate
9	5/25/21 7:26:33	5/25/21 7:28:45	anonymous		Sushi;	563456	Hannah
10	5/25/21 7:26:27	5/25/21 7:29:01	anonymous		Sushi;	67743	Brian

In Power BI Desktop, select **Import data from Excel** and choose your Excel file.

Select the required table, then click **Load**.



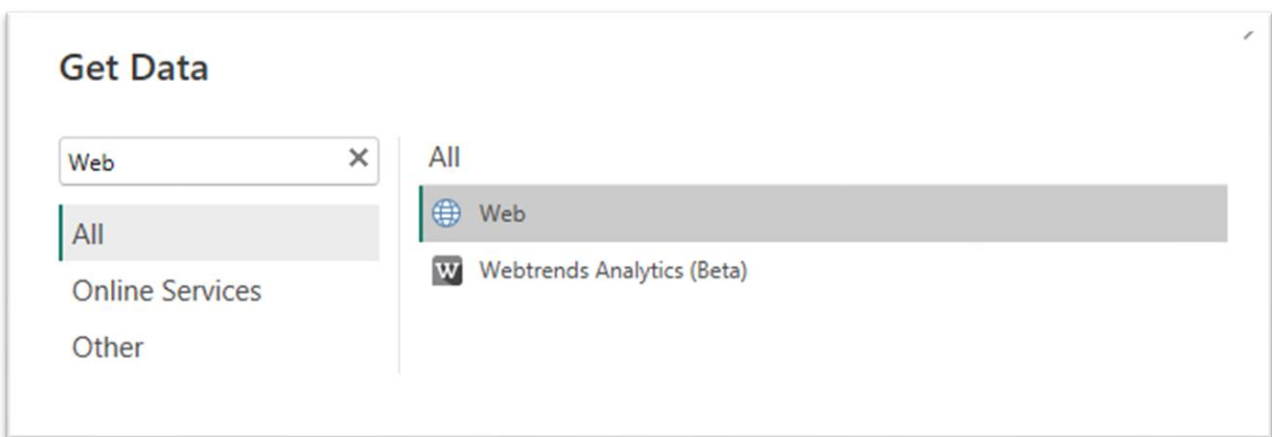
Below, we're using a column chart to detail the results of our survey.



Another option worth mentioning if you want this to be automated, is to create a Power Automate flow that triggers each time a survey response is received, that then copies the response into a SharePoint list. Then you base your Power BI reports on that SharePoint list data.

Getting data from the web

You can import data from the web just like we did with SharePoint. Search for **web**, select it and click **Connect**.



For this example, we're going to use a Wiki page entry on the bestselling automobiles:

https://en.wikipedia.org/wiki/List_of_best-selling_automobiles

When prompted, insert the URL. Next, it brings up many tables that we can import.

However, we're sticking to just one – Brand bestsellers[Edit].

Navigator

Display Options

HTML Tables [8]

☐ Table 1
☐ Overview of globally best-selling auto...
☐ Sortable and collapsible table
☒ Brand bestsellers[edit]
☐ Class bestsellers[edit]
☐ Fastest-selling[edit]
☐ Table 2
☐ Table 3

Suggested Tables [8]

☐ Table 4
☐ Table 5
☐ Table 6
☐ Table 7
☐ Table 8
☐ Table 9
☐ Table 10
☐ Table 11

Text [2]

☐ HTML Code

Add Table Using Examples

Table View

Web View

Brand bestsellers[edit]

Column1	Column2
Brand	Image
Alfa Romeo	1979 Alfasud ti
Alpine	Alpine-Renault A310
American Motors	AMC Hornet.
Aston Martin	Aston Martin DBX.
Audi	
Autobianchi	1973 Autobianchi A112 E.
Baojun	Baojun 510
BMW	First generation BMW 3 Series coupé.
Buick	1964 Buick LeSabre.
Bugatti	1929 Bugatti Typ 40 Grand Sport Tourer.
BYD	
Cadillac	Cadillac De Ville.
Checker	Checker Marathon.
Chevrolet	1965 Chevrolet Impala.
Chrysler	1961 Chrysler Newport.
Citroën	First generation Citroën 2CV.
Continental	Continental Mark II.
Crosley	1948 Crosley.
Dacia	Dacia Sandero.
DeSoto	Dodge DeSoto Custom.

Load

Transform Data

Cancel

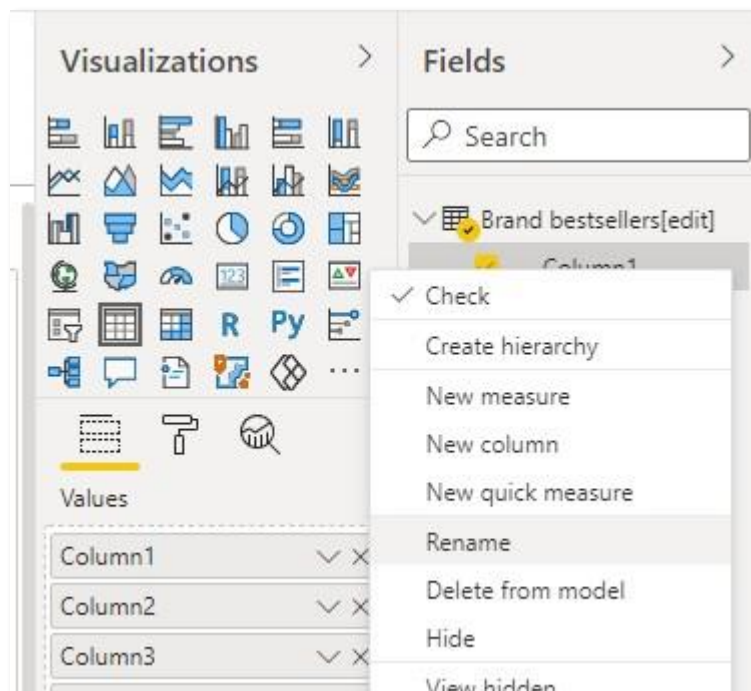
However, once we bring in the data, we must ensure that the columns are correctly labelled.

We're going to rename and/or remove columns to showcase the following data:

- Brand
- Automobile
- Production
- Units Sold

We're going to remove the column for Notes.

Under fields, select each column and rename it in line with the original column name in Wikipedia.

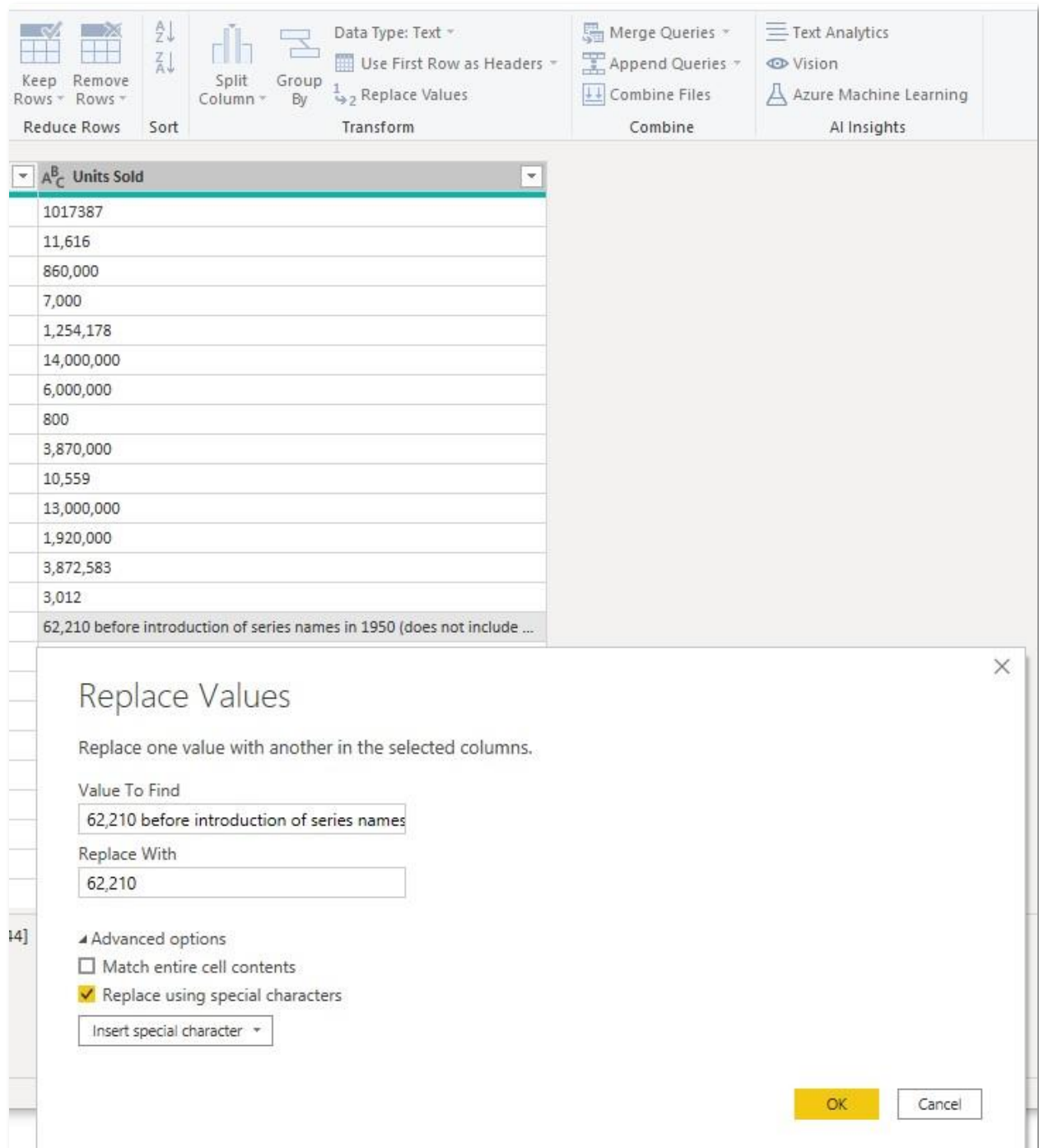


Once arranged, our data looks like the following.

Brand	Automobile	Production	Units Sold
Alfa Romeo	Alfa Romeo Alfasud	1972–1989	1,017,387.[40]
Alpine	Alpine A110 (2017)	2017–present	18,985.[41][42]
American Motors	AMC Hornet	1970–1977	Approximately 860,000.[43][44]
Aston Martin	Aston Martin DBX	2020–present	10,675 to end of 2023.[45][46][47][48]
Audi	Audi A4	1994–present	Over 7,500,000 to 2019[49]
Autobianchi	Autobianchi A112	1969–1986	1,254,178.[50]
Baojun	Baojun 510	2017–present	972,042 to 2021.[51]
BMW	BMW 3 Series	1975–present	Over 14,000,000 to 2015.
Brand	Automobile	Production	Units sold
Bugatti	Bugatti Type 40	1926–1930	Approximately 800.[54]
Buick	Buick LeSabre	1959–2005	Over 6,000,000.[53]
BYD	BYD F3	2005–2022	1,684,904, excluding exports.[55]
Cadillac	Cadillac De Ville	1959–2005	Approximately 3,870,000 excluding early Series 62 hardtops, 1961–1963, 1970–1971.
Checker	Checker Marathon	1961–1982	10,559 not counting taxicabs and private sales for 1961–63, 1970–1971.
Chevrolet	Chevrolet Impala	1958–2020	Over 13,000,000 to 1996.[56]
Chrysler	Chrysler Newport	1961–1981	Approximately 1,920,000 (excluding early Newport hardtops).[43]
Citroën	Citroën 2CV	1948–1990	3,872,583. Including commercial variants, the total figure is approximately 4,000,000.
Continental	Continental Mark II	1956–1957	3,012 (only car produced by the short lived Continental division).

As the data in the Units Sold column is unstructured – it contains numbers and words – we’re going to clean it up by selecting **Transform data** under the **Home** tab and Replace Values for

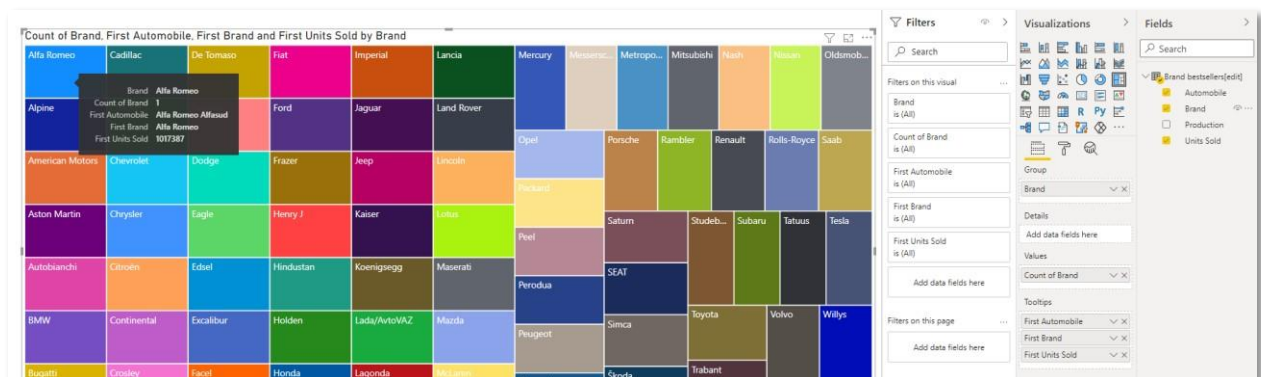
some of the items in that column. We want to remove references to footnotes and other miscellaneous text. **Right click the column** in the Power Query editor and select **Replace Values**. (You will have to do this manually for each item. You could also attempt to use Copilot or ChatGPT to handle this for you and replace the record fields as necessary).



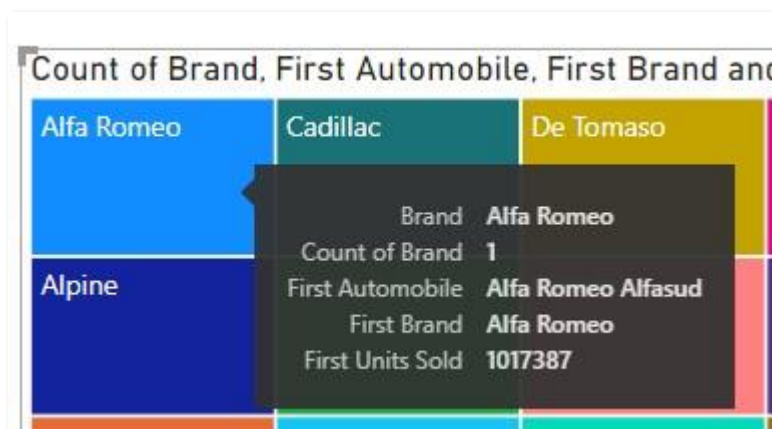
Next, make sure that your values are in the order that you want them to appear. You don't have to number them, but we have; what's important is that they appear in the order that you want them to appear in.



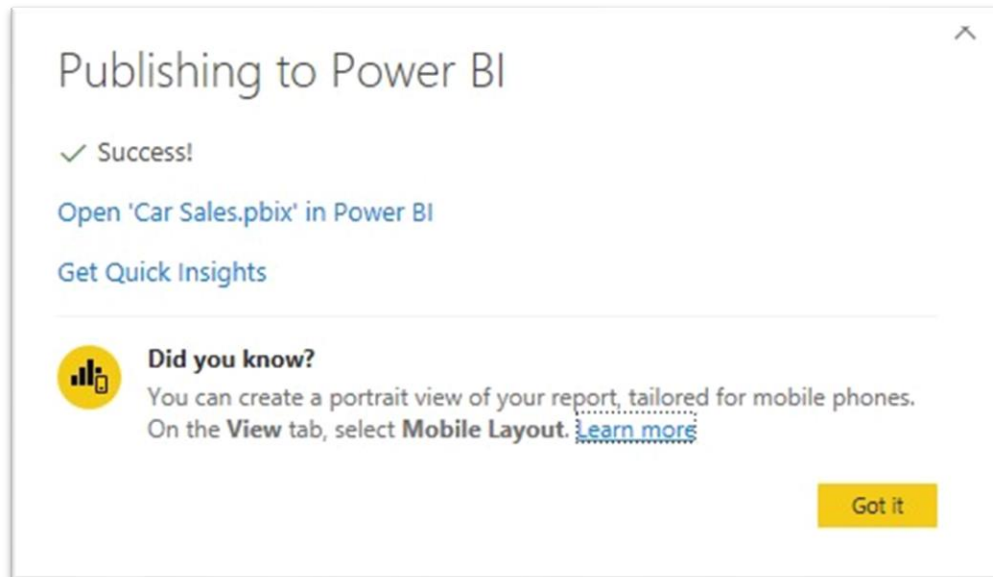
Next, we're going to turn our data in a Treemap. When we select the Treemap visualization, our data is displayed in boxes.



When we hover over a box, we get the data relating to each car brand.



Once we have our data saved, we're going to **Publish** it to a workspace in the Power BI service by selecting **Publish** and choosing a workspace that we've created called Car Sales.



Getting data from other sources

There are many different data sources that you can connect to, and you can connect to multiple data sources at the same time.

Here is a list of file sources that you can connect to and experiment with:

- Excel
- Text/CSV
- XML
- JSON
- Folder
- PDF
- Parquet
- SharePoint folder

Here is a list of all databases that you can connect to:

- SQL Server database
- Access database
- SQL Server Analysis Services database
- Oracle database
- IBM Db2 database
- IBM Informix database (Beta)
- IBM Netezza
- MySQL database
- PostgreSQL database
- Sybase database
- Teradata database
- SAP HANA database
- SAP Business Warehouse Application Server
- SAP Business Warehouse Message Server
- Amazon Redshift
- Impala
- Google BigQuery
- Google BigQuery (Microsoft Entra ID)(Beta)
- Vertica
- Snowflake
- Essbase
- AtScale Models
- Actian (Beta)
- Amazon Athena
- AtScale cubes
- BI Connector
- Data Virtuality LDW
- Denodo
- Dremio Software
- Dremio Cloud
- Exasol
- Indexima
- InterSystems IRIS (Beta)
- Jethro (Beta)
- Kylligence
- Linkar PICK Style / MultiValue Databases (Beta)
- MariaDB
- MarkLogic

- MongoDB Atlas SQL (Beta)
- TIBCO® Data Virtualization
- Exact Online Premium (Beta)

Data refresh

Depending on how you connect your data, it will refresh differently.

According to Microsoft, 'A Power BI refresh operation can consist of multiple refresh types, including data refresh, OneDrive refresh, refresh of query caches, tile refresh, and refresh of report visuals.'

Here is a quick guide:

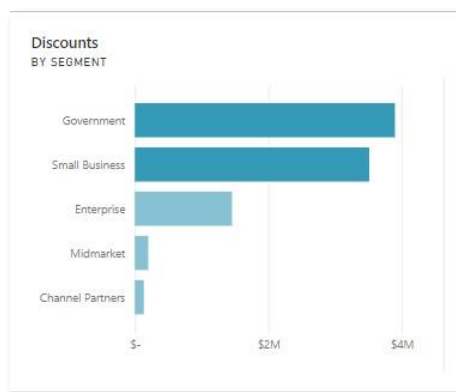
Storage mode	Data refresh	OneDrive refresh	Query caches	Tile refresh	Report visuals
Import	Scheduled and on-demand	Yes, for connected semantic models	If enabled on Premium capacity	Automatically and on-demand	No
DirectQuery	Not applicable	Yes, for connected semantic models	Not applicable	Automatically and on-demand	No
LiveConnect	Not applicable	Yes, for connected semantic models	Not applicable	Automatically and on-demand	Yes
Push	Not applicable	Not applicable	Not practical	Automatically and on-demand	No
Table source and more information, see here https://docs.microsoft.com/en-us/power-bi/connect-data/refresh-data					

The top ten visualizations

There are many visualizations that come out-of-the-box with Power BI. Also, there are many being created by third parties, too. **Here are our top ten...**

Bar charts

Call us old fashioned, but there's something elegant and straight forward about a simple bar chart.



Donut charts

Another classic, and for a very good reason – it gives you a clear, unambiguous view of who is devouring most of the data.



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-doughnut-chartsbi/visuals/power-bi-visualization-doughnut-charts>

Scatter charts

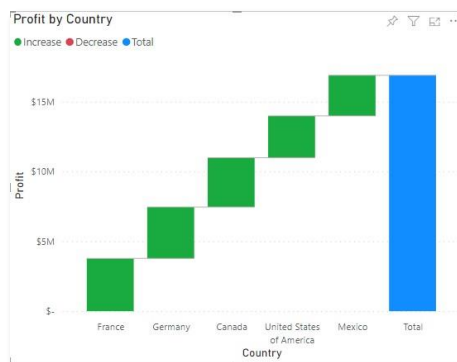
Dots floating around in clear, clean space presents data in an original and simple way.



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-scatterbi/visuals/power-bi-visualization-scatter>

Waterfall chart

Although the waterfall may at times look like something out of Minecraft, it has a certain charm – and solid data presentation.



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-waterfall-chartsbi/visuals/power-bi-visualization-waterfall-charts>

Maps

It's hard to get a better way of reporting country-by-country data than a map.



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/desktop-shape-mapbi/visuals/desktop-shape-map>

Tree maps

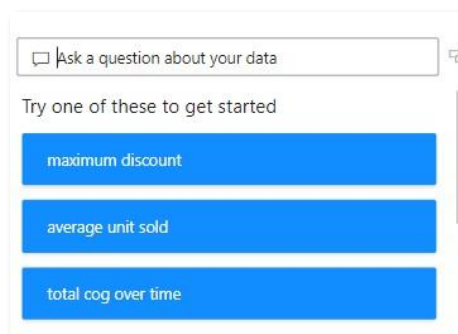
What could be nice than an interactive wall of beautifully colour data bricks?



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-treemapsbi/visuals/power-bi-visualization-treemaps>

Q&A

Driven by AI, this does exactly what it's called – query your data as if it's your very own data analyst.



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-q-and-abi/visuals/power-bi-visualization-q-and-a>

Funnel chart

Another simple and effect way of presenting data – check what's in your linear funnel.



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-funnel-chartsbi/visuals/power-bi-visualization-funnel-charts>

Gauge

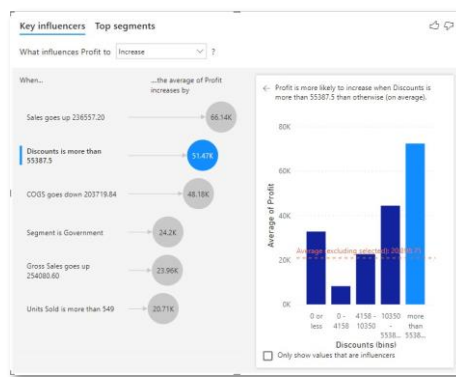
What's more visual than a rev-counter data visual that displays your data against a targeted goal?



For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-radial-gauge-chartsbi/visuals/power-bi-visualization-radial-gauge-charts>

Key influencers

Last, but not least, this one of the finest AI driven visualizations on the market today that allows you to Q&A your data – more on this in the next chapter of this eBook.



For more about visualizations, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-types-for-reports-and-q-and-abi/visuals/power-bi-visualization-types-for-reports-and-q-and-a>

Furthermore, for other Microsoft visualizations and those which have been created by third parties, see here <https://appsource.microsoft.com/en-US/marketplace/apps?product=power-bi-visuals>

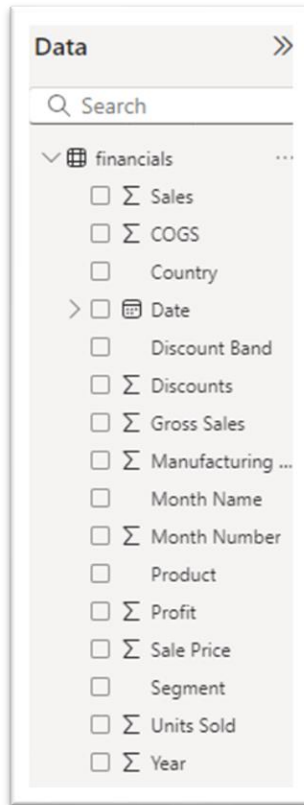
Key influencers visualizations

Key influencers are an AI-driven visualization which tries to drill down into your data in order to figure out the key reasons that are influencing aspects of your data. In summary, it tries to understand the reasons why your KPIs are moving in one direction or another.

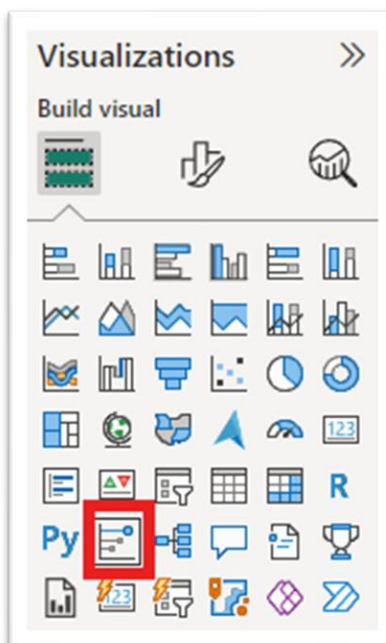
Please note: Key influencers can be used for both the Power BI service and the Power BI Desktop. In this example, we're using Power BI Desktop for a better experience.

In Power BI Desktop, select Get data and choose a file. We're selecting our Financial Sample Excel file and one specific sheet.

Once loaded, you can see the columns from the Excel sheet.



Select the Key influencers visualization.



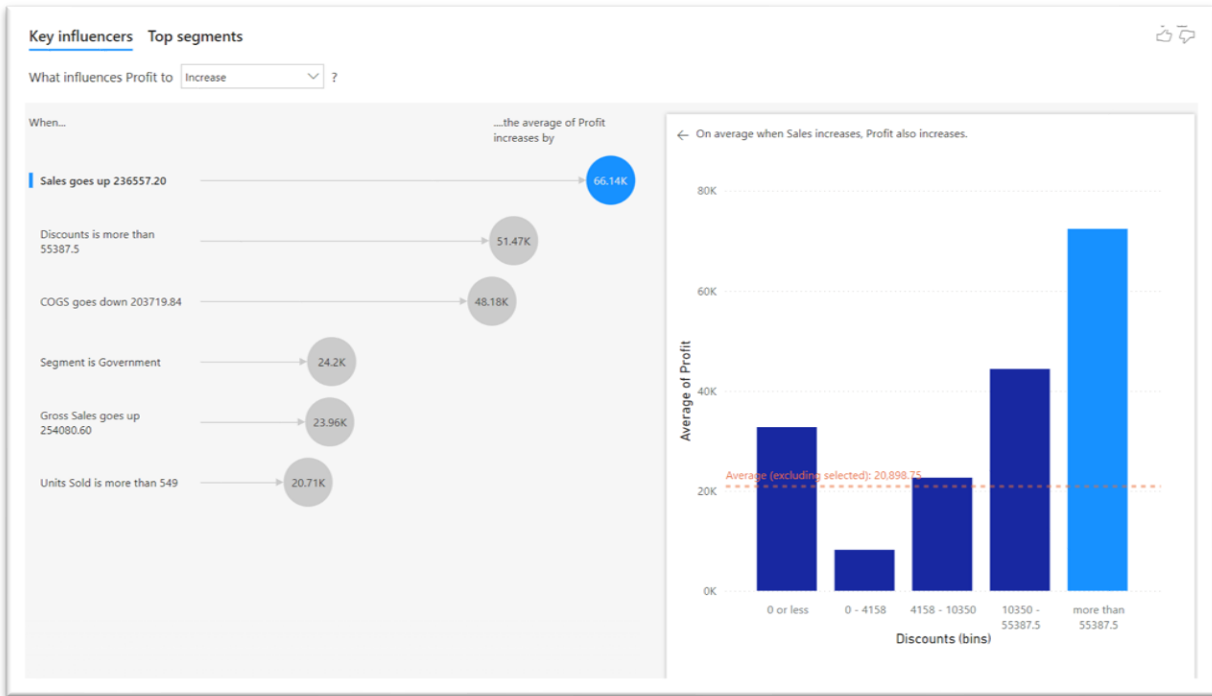
When you select the Key influencers visualization, you must input fields in the Analyze section and then add more fields to the Explain by section. In our example, we have added Profit to the Analyze section; and we have put every other field in the Explain by section.

Analyze	
Profit	▼ X

Explain by	
Year	▼ X
Units Sold	▼ X
Segment	▼ X
Product	▼ X
Month Number	▼ X
Month Name	▼ X
Gross Sales	▼ X
Manufacturing Price	▼ X
Discounts	▼ X
Discount Band	▼ X
Country	▼ X
COGS	▼ X
Sales	▼ X

The above settings return the following:

- Key influencers show us the top contributing factors to the selected item.
- Top segment shows the top segments that contribute to the selected item.
- In our example, the dropdown box is a binary choice between Increase and Decrease.
- On the right hand-side we see a graphic which is a visual interpretation.
- The left pane contains the actual list of key influencers.

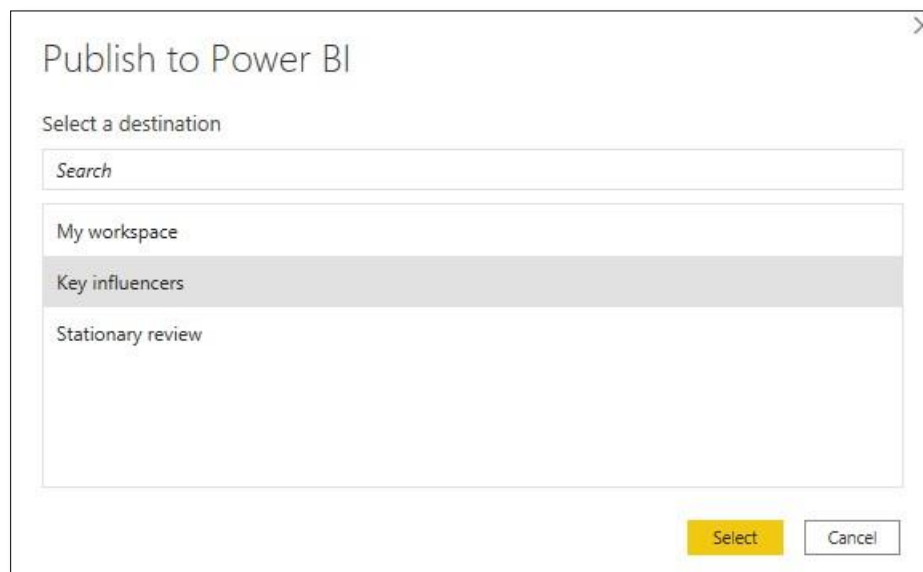


We can also filter the data by selecting Filters. In the below example, we have filtered our data to only include references to Canada and France, and the visualizations adjust accordingly.



Now that we have our key influencers report in Power BI Desktop, let's save it to the Power BI service. We have created a workspace called Key influencers; this is going to be the destination for the above report.

On the Home section of the ribbon, select Publish. We're given a choice as to where we'll publish to. In our case, we're selecting Key influencers.



Once published to the Power Bi service, you have the option to get quick insights or to further share within others as a link or in in Teams.

For more information, see here <https://docs.microsoft.com/en-us/powerhttps://docs.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-influencersbi/visuals/power-bi-visualization-influencers>

Dashboards

A Power BI dashboard is a single page which contains several visualizations. When used well, they give the viewer a high-level summary of an organisation's activity.

A dashboard is based on data which comes from reports. In essence, a dashboard can contain multiple visualizations of data from many reports.

While a dashboard may sound like a report, there are differences.

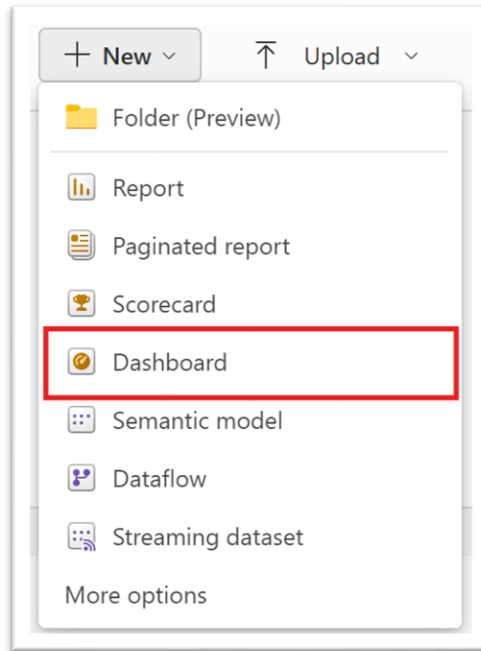
Capability	Dashboards	Reports
Pages	One page	One or more pages
Data sources	One or more reports and one or more datasets per dashboard.	A single dataset per report
Drilling down in visuals	Only if you pin an entire report page to a dashboard.	Yes
Available in Power BI Desktop	No	Yes. Can build and view reports in Power BI Desktop.

Filtering	No. Can't filter or slice a dashboard. Can filter a dashboard	Yes. Many different ways to filter, highlight, and slice.
	tile in focus mode , but can't save the filter.	
Featured	Yes. Can set one dashboard as your featured dashboard.	No
Favourite	Yes. Can set multiple dashboards as favourites.	Yes. Can set multiple reports as favourites.
Natural language queries (Q&A)	Yes	Yes, provided you have edit permissions for the report and underlying dataset.
Set alerts	Yes. Available for dashboard tiles in certain circumstances.	No

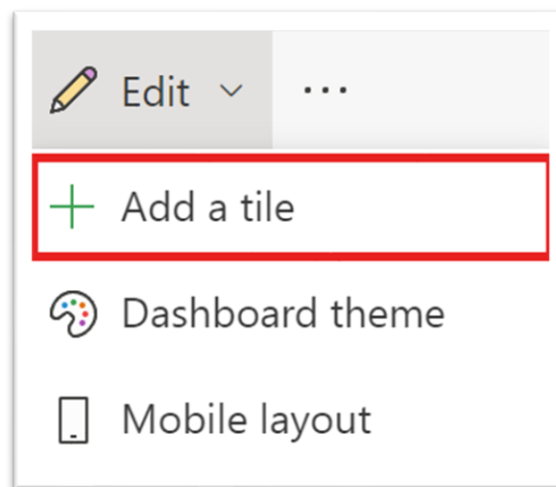
Subscribe	Yes. Can subscribe to a dashboard.	Yes. Can subscribe to a report page.
Can see underlying dataset tables and fields	No. Can export data but can't see tables and fields in the dashboard itself.	Yes
Table source: https://docs.microsoft.com/en-us/power-bi/create-reports/service-dashboards		

Creating a dashboard

While in your workspace (in our example, our workspace is EbookWorkspace) select **+New** and **Dashboard**. We're going to name it *World View*.



Next, we're going to add a tile by selecting **Edit** → **Add a tile**.

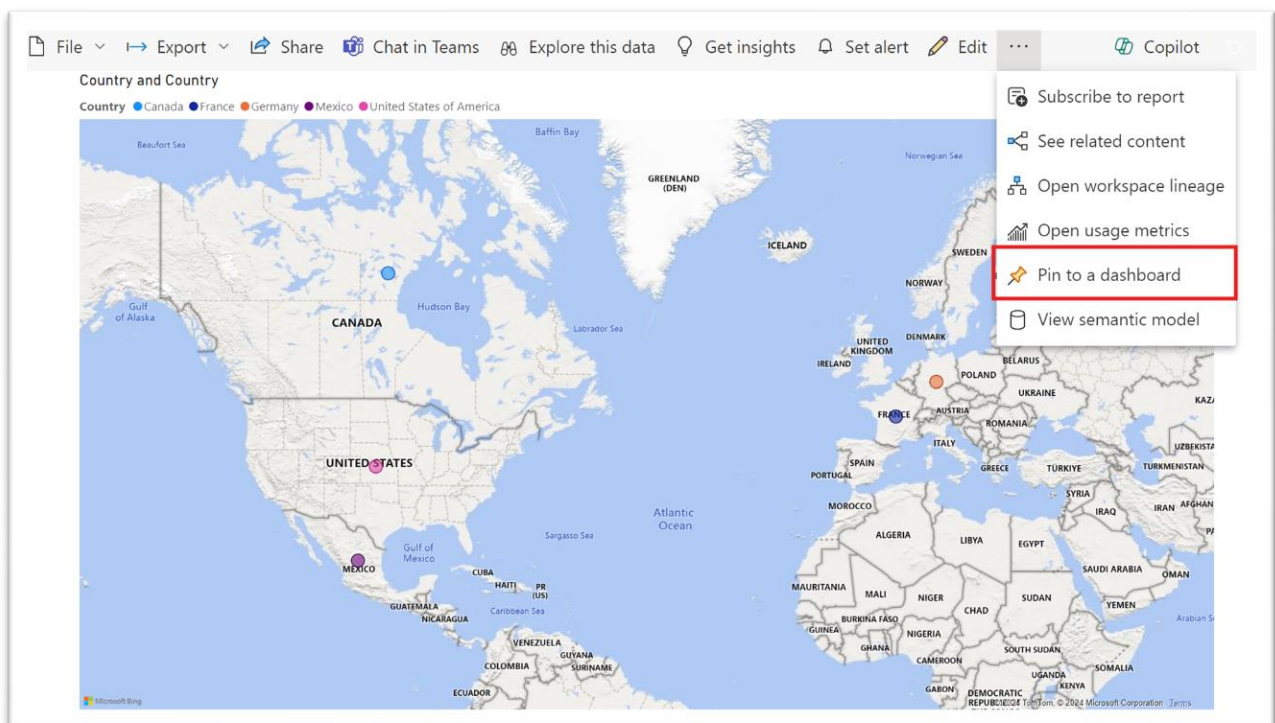


We are given the following choices:

- Web content
- Image
- Text box
- Video
- Read-Time Data

Let's select Text box. We're going to title our dashboard ACME Corporation – Global Sales. Within the pane, you can resize and change the font too.

Next, we're going to bring in the data map from our previous example. Open the data map and select the Pin to a dashboard icon (You may need to expand the options on your toolbar using the ellipsis).



When prompted, choose Existing dashboard and in our example, we're pinning it to World View.

×

Preview: Last saved state

FinancialSampleReport

PAGE 1



Pin to dashboard

Select an existing dashboard or create a new one.

Where would you like to pin to?

☒ Existing dashboard
 ☐ New dashboard

Select existing dashboard

World View

i Pin live page enables changes to reports to appear in the dashboard tile when the page is refreshed.

Pin live

Cancel

We now have two tiles on our dashboard.

Ask a question about your data

ACME Corporation - Global Sales

Units Sold by Country and Country

Country
 Canada
France
United States of America
Mexico
Germany



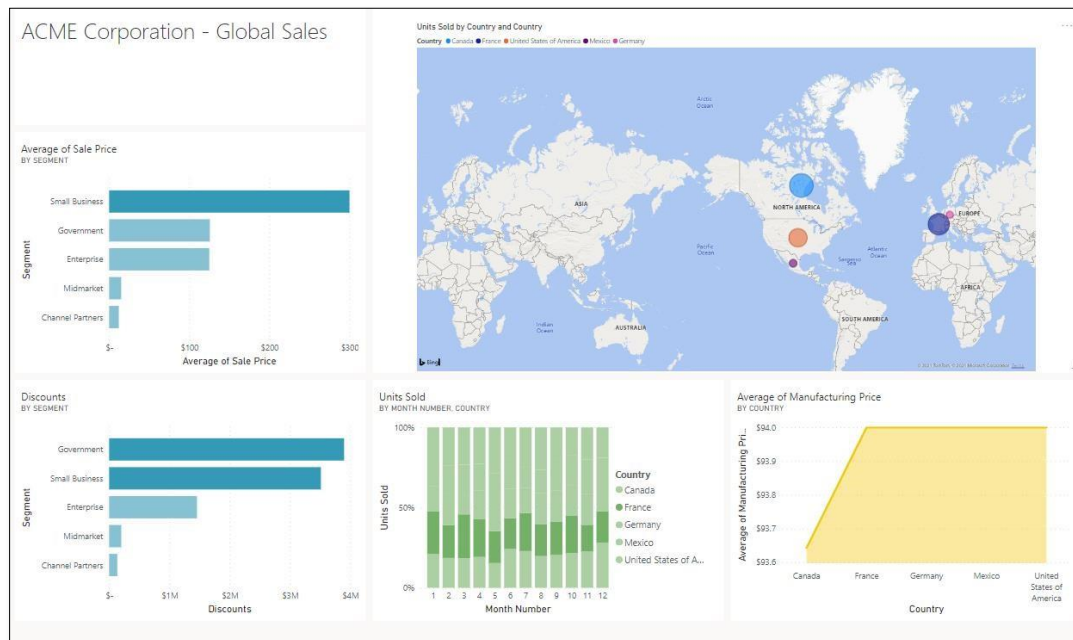
68

collab365.

Let's add some tiles based on the Excel file Financial Sample and quick insights.

We've added:

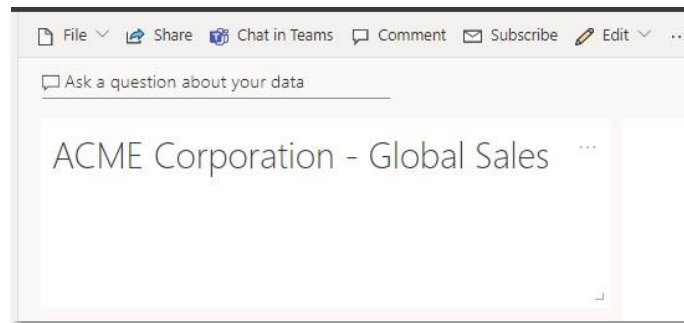
- Average of Sale Price
- Discounts
- Units Sold
- Average of manufacturing price



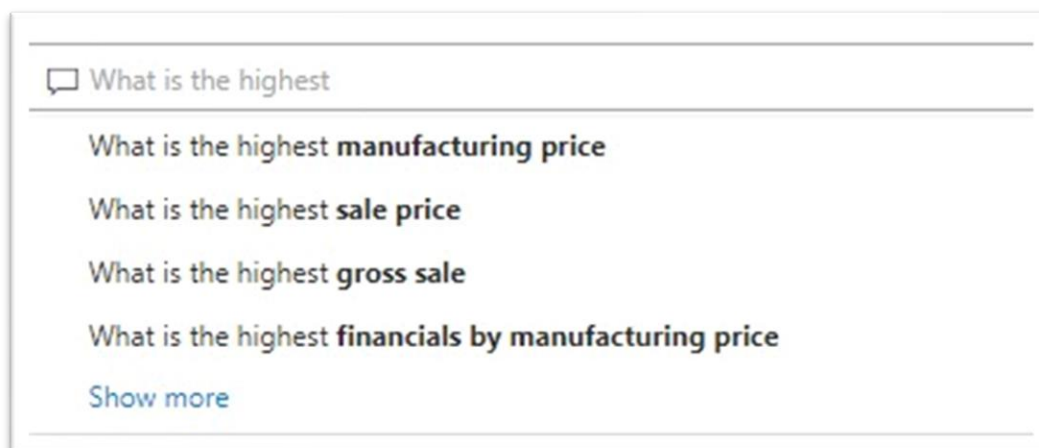
You can share your final Dashboard using the same methods expressed previously in this eBook.

Ask a question about your data

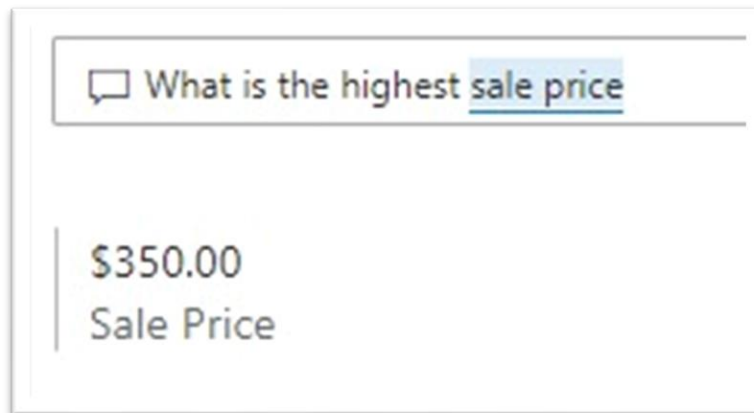
At the top of our dashboard – and on other visualizations too – you will be prompted to Ask a question about your data.



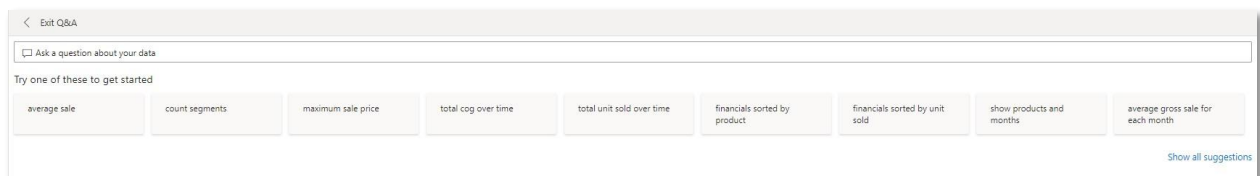
It helps if you have some knowledge of how your data is structured in terms of what columns exist as basically, when you ask a question - and hope to get an answer – you are really comparing data in one column to another or asking binary questions such as what the highest or lowest price of data in another column is. If we start asking the following question What is the... we get these options:



If we select What is the highest sale price, the answer appears:



Furthermore, there are predefined questions which act as a starting point. By default, our predefined questions cover the average sale, count segments, maximum sale price and many more.



Creating an App

Power BI allows you to create apps. When you think of a Power BI app, think of it as being made up of reports and dashboards. The purpose of creating an app is to share it with a broad audience within your organisation. When you share an app, users can:

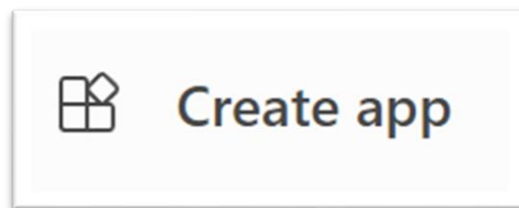
- Look but cannot modify its content
- View it in the Power BI service a mobile app
- Get updated automatically or whenever you decide to update them

For more information on apps and how the licencing works, see here

<https://docs.microsoft.com/en-us/power-bi/collaborate-share/service-create-distribute-apps>

So, let's create our first app...

As apps are created in workspaces, let's open our EbookWorkspace and select **Create App** in the top right of the screen.



Give your app a **name** and a **description**. You can give the app a colour, too.

1 Setup
2 Content*
3 Audience*

Build your app

App name *
WorkspaceApp

Description *
This is a description
179 characters left

App logo
 Upload
Delete

App theme color


Contact information
☒ Show app publisher
☐ Show items contacts from the workspace
☐ Show specific individuals or groups

Advanced settings

Next: Add content

Next, we can add any of the content that exists in our Workspace to it by selecting the **Add content** button. Select everything you want to add and click **Add**.

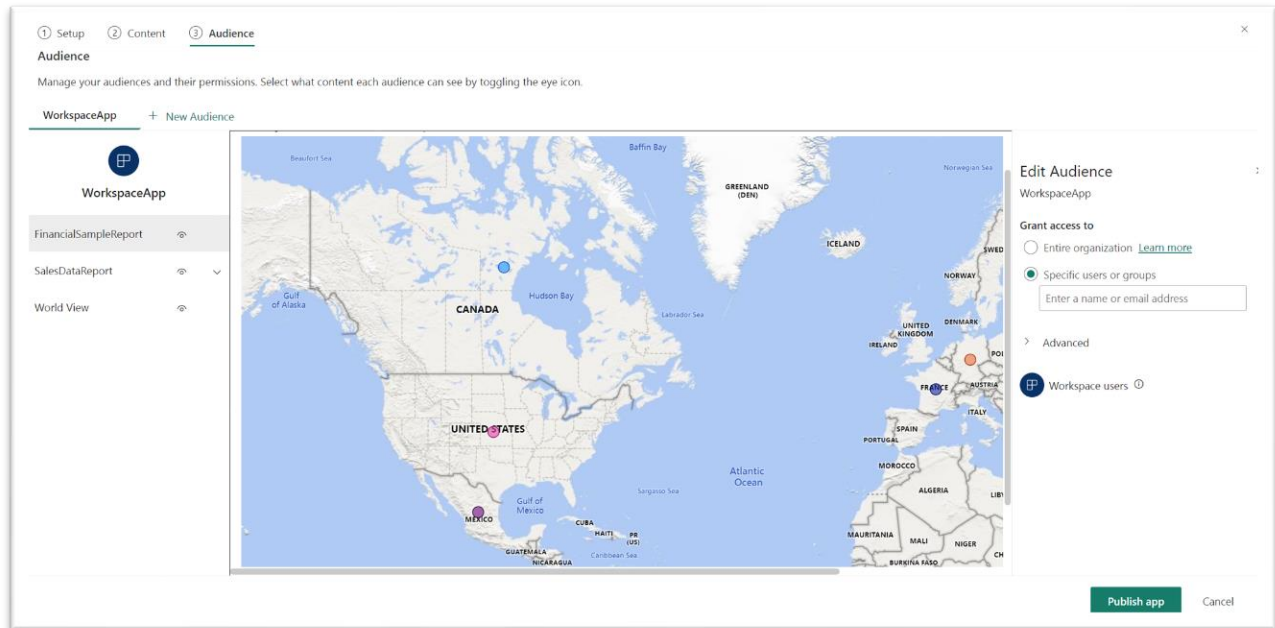
Add content

Insert reports, dashboards, and workbooks directly from your workspace. You can also add website links.

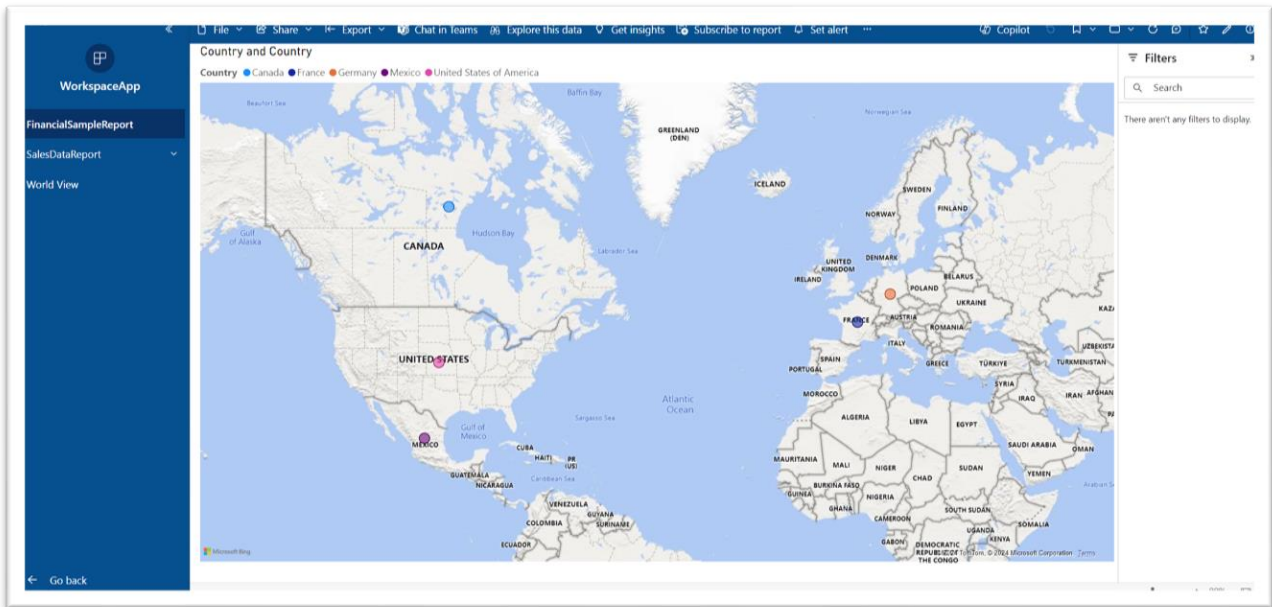
Add from workspace
Add a link

☒		Name	Type	Owner	Refreshed
☒		FinancialSampleReport	Report	EbookWorkspace	09/07/24, 13:55:35
☒		SalesDataReport	Report	EbookWorkspace	08/07/24, 14:29:50
☒		World View	Dashboard	EbookWorkspace	—

We can now create new audiences that have access to specific pieces of data, semantic models, reports and dashboards and you can configure as necessary by using the on-screen controls to set audience access.



When you are happy, select **Publish App**. This process may take between 5 and 10 minutes before you can view your app, as security is being applied to your workspace and data. You'll then be able to **Go to app**, where you'll be able to view the published application.



From here, you can share your app in Teams & SharePoint, giving your users the flexibility of interacting with your dashboards and reports with a clean view and user friendly navigation.

How is Power BI Licensed?

Now you have had experience of some of the capabilities of Power BI, you'll know it is a wide-ranging subject with many avenues to explore.

We haven't touched on many of these (e.g. DAX language, data modelling and more) in favour of giving you a sample to what's on offer.

One thing we do not want to leave out however is again, just an introduction to the world of Power BI licences.

There are three main kinds of Power BI service per-user licenses at the time of writing: Free, Pro and Premium Per User.

Important note: Licencing often Changes. Whilst the below is a rough guide, it's always worth checking the links or searching for Power BI pricing for the most up to date information)

License type	Capabilities when workspace is in shared capacity	Additional capabilities when workspace is in Premium capacity
Power BI (free)	Access to content in My Workspace. Please note you cannot share content with this license	Consume content shared with them
Power BI Pro	Publish content to other workspaces, share dashboards, subscribe to dashboards and reports, share with users who have a Pro license	Distribute content to users who have free licenses
Power BI Premium Per User	Publish content to other workspaces, share dashboards, subscribe to dashboards and reports, share with users who have a Premium Per User license	Distribute content to users who have free and Pro licenses
Table source: https://docs.microsoft.com/en-us/power-bi/fundamentals/service-features-license-type		

And here are some of the differences within the Power BI Pro and premium plans.

It is worth noting, the **position on licences changes regularly** so it is worth finding a trusted advisor or partner for your licence questions if you plan to use Power BI a lot.

Note: according to Microsoft, a capacity ‘is a dedicated set of resources reserved for exclusive use. It enables you to publish dashboards, reports, and datasets to users, without having to purchase per-user licenses.’

Feature	Free Account	Power BI Pro	Power BI Premium Per user	Power BI Embedded
Create reports with desktop	✓	✓	✓	✓
Publish reports to share and collaborate		✓	✓	✓
Advanced AI			✓	✓
Advanced Dataflows			✓	✓
Advanced Datamarts			✓	✓
XMLA endpoint read/write			✓	✓
Consume Power BI content without a paid per-user license				✓
Access to Fabric workloads				
Copilot in Fabric				

Model memory size limit		1 GB	Varies	Varies
Refresh rate for Power BI datasets		8/day	48/day	48/day
Maximum storage (native storage)		10 GB/user	100 TB	Varies
Data security and encryption	✓	✓	✓	✓
For the most up to date information including costs, visit the table source here: https://powerbi.microsoft.com/en-us/pricing/				

Next steps

In this eBook we have touched on just some of the most powerful features that Power BI has to offer. However, as it's a very powerful and wide-ranging tool, the examples which we have covered are only a starting point for your Power BI journey.

For example, you can learn more about:

- Datasets: For more information, see here <https://docs.microsoft.com/en-us/powerbi/connect-data/service-datasets-hub>
- Goals: For more information, see here <https://powerbi.microsoft.com/en-us/blog/introducing-goals-in-power-bi/>
- Apps: There are many third-party apps which can connect to your version of Power BI. These apps simply plug into Power BI in a similar way that apps plug into Teams. For more information, see here <https://powerbi.microsoft.com/en-us/blog/distribute-to-large-audiences-with-power-bi-apps/>
- Deployment pipelines: This feature allows you to manage and control the lifecycle of your data and is based on three stages: The Deployment phase; the Test phase, and the Production phase. For more information, see here https://docs.microsoft.com/en-us/power-bi/create-reports/deployment-pipelines-overview?WT.mc_id=welcome_page

Our recommendation is to take the next step and go explore what we have shared in the Academy as well as taking part in a workshop to begin to build your knowledge.